

Estimating properties of semi-resolved globular clusters in M31: A forward modelling and deep learning approach

Arn Marklund¹, Paolo Bianchini¹, Paul Magron², Abbas Askar³, and Ariane Lançon¹

¹ Université de Strasbourg, CNRS, Observatoire astronomique de Strasbourg, UMR 7550, F-67000 Strasbourg, France
e-mail: arn.marklund@astro.unistra.fr

² Université de Lorraine, CNRS, Inria, LORIA, F-54000 Nancy, France

³ Nicolaus Copernicus Astronomical Center, Polish Academy of Sciences, ul. Bartycka 18, 00-716 Warsaw, Poland

Received September 15, 1996; accepted March 16, 1997

ABSTRACT

Recent and upcoming wide-field photometric surveys will observe hundreds of thousands of globular clusters (GCs) in the next decade. This poses an emerging challenge: to derive reliable estimates of their properties using methods that scale to large volumes of data. We present a new framework that combines forward modelling with deep learning, allowing accurate and detailed analysis of GCs. We develop CREMANT, a forward-modelling code that is used to create realistic mock images of GCs from state-of-the-art N -body simulations, and embed them in real backgrounds. We generate mock images resembling real HST observations of M31 (PHAT and PHAST surveys) and train a neural network, π -DOC, to decontaminate GC images from field stars, reconstruct 2D mass maps, and infer global properties (age, distance, ellipticity, and position angle) from multi-band photometric images and pixel colour-magnitude diagrams.

We find that on synthetic data, π -DOC constrains cluster mass and age to within $^{+0.23}_{-0.50}$ dex and $^{+2.86}_{-3.46}$ Gyr at 2σ — a conservative confidence range that remains comparable to typical M31 literature uncertainties and only $\sim 1.5\times$ wider than typical mass uncertainties for fully resolved Galactic GCs. Decontamination remains robust even at low signal-to-noise, suggesting π -DOC could be re-purposed for simultaneous GC detection. Distance ($^{+77}_{-80}$ kpc) and ellipticity ($^{+0.06}_{-0.09}$) estimates are comparatively more uncertain, with the latter primarily exacerbated by faint and only mildly elliptical GCs.

Applied to GCs in the PHAT and PHAST footprints, the model shows agreement with the existing literature within a factor of four for total masses, while efficiently scaling to large volumes of data with an inference speed of ~ 10 objects per second (full M31 dataset in ~ 30 s). We release a publicly available catalogue of π -DOC's estimates for 349 GCs in M31, as well as both codes for CREMANT and π -DOC. This work illustrates that the combination of forward modelling and deep learning is a scalable and physically motivated approach for inferring GC properties from large multi-band photometric datasets.

Key words. methods: numerical, deep learning – globular clusters: general – galaxies: star clusters: general – objects: M31

1. Introduction

Globular clusters (GCs) are dense, gravitationally bound stellar systems composed of up to a few million stars. They are among the oldest known stellar populations in the Universe, with Galactic GCs showing ages of up to 13 Gyr (Leaman et al. 2013; Valcin et al. 2025, 2026). Thanks to recent gravitationally lensed observations with the James Webb Space Telescope (JWST), the progenitors of present-day GCs have been identified at redshifts corresponding to only about 500 Myr after the Big Bang (Vanzella et al. 2023; Adamo et al. 2024; Mowla et al. 2024). While their exact origin still remains an open question, their formation is linked to key phases of structure formation in the early Universe (e.g. high pressure star formation and galactic merger events Krujssens 2015; Lahén et al. 2020, 2025; Taylor et al. 2025).

Over their roughly 13 Gyr of evolution, GCs are shaped by internal dynamical processes, such as two-body relaxation and stellar evolution, as well as by external effects, such as the coupling to the tidal field of their host galaxy. Related to the possible formation channel of GCs during a galaxy's assembly phase, some GCs are also thought to have been accreted during galaxy merger events (Forbes & Bridges 2010; Renaud et al. 2017; Forbes et al. 2018; Forbes 2020; Massari et al. 2019; Myeong

et al. 2019; Pfeffer et al. 2021; Chen & Gnedin 2024), or even be the remnant nuclei of dwarf galaxies (e.g. Ibata et al. 1995; Alfaro-Cuello et al. 2019; Pagnini et al. 2025, 2026). The interplay between these mechanisms and the resulting complex evolution over a Hubble-time, make GCs useful testbeds for many different astrophysical topics: stellar evolution models, dynamical theories of collisional systems, and the long-term effects of galactic environments on stellar populations.

The development of high-precision N -body (Aarseth 2003) and Monte Carlo codes (such as the MOCCA and CMC codes Hypki & Giersz 2013; Giersz et al. 2013; Rodriguez et al. 2016), and particularly GPU-accelerated implementations such as Nbody6++GPU (Wang et al. 2015), has made it possible to simulate clusters containing up to and above a million stars, reaching parity with the mass regime of typical Galactic GCs (e.g. Arca sedda et al. 2024; Wang et al. 2016; Bianchini et al. 2026). These modern simulations also include ingredients such as time-dependent tidal fields (Renaud et al. 2011; Webb et al. 2024) and stellar evolution (Kamlah et al. 2022; Bissekennov et al. 2025; Wu et al. 2026), offering increasingly realistic representations of GC dynamics, their stellar remnant populations, and internal rotation. Despite these advances, such direct simulations remain extremely demanding in terms of computational time and energy

consumption, limiting their use for large-scale parameter studies or statistical comparisons (Bianchini et al. 2026).

In parallel, the growth of deep-field, as well as wide-field surveys, are pushing us into a new era for GC observations. Space- and ground-based facilities such as the Hubble space telescope (HST), the James Webb Space Telescope (JWST Gardner et al. 2023), Euclid (Euclid Collaboration et al. 2022), and the Vera C. Rubin Observatory (LSST Ivezić et al. 2019) are rapidly increasing both the number of known clusters and the redshift range over which they can be studied. Recent projections estimate that Euclid alone could detect several hundred thousand GCs across the local Universe (Euclid Collaboration et al. 2025). Upcoming missions, including the Roman space telescope (Mosby et al. 2020), are expected to continue this trend by delivering deep, high-resolution multi-band imaging over large areas of the sky. These developments underline an emerging challenge: the need for scalable, automated tools capable of analysing vast samples of clusters efficiently.

Determining fundamental GC properties such as total mass and age remains non-trivial, in particular due to their complex dynamical evolution (Spitzer 1987; Vesperini & Heggie 1997; Baumgardt & Makino 2003; Gieles et al. 2011; Trenti & van der Marel 2013; Watkins et al. 2015; Bianchini et al. 2016; Kuzma et al. 2016; Webb & Vesperini 2016; Malhan et al. 2018; Jindal et al. 2019) and because of degeneracies in simple stellar population modelling, notably the degeneracies between age, metallicity, and extinction (Worthey 1994, 1999; Usher et al. 2019, 2024) and systematic differences between simple stellar population (SSP) models (Maraston 2005; Conroy et al. 2009; Conroy & Gunn 2010; Fan & de Grijs 2012).

Typical approaches involve (i) assuming a fixed mass-to-light (M/L) ratio, which is not strictly valid for GCs (as demonstrated in e.g. Baumgardt 2017; Bianchini et al. 2017), (ii) deriving dynamical masses from kinematic data (Bellini et al. 2017), which becomes increasingly difficult at extragalactic distances, or (iii) constructing large grids of N -body simulations (Baumgardt 2017), a procedure that is both computationally expensive and environmentally costly (Bianchini et al. 2026). Monte-Carlo based methods, such as MOCCA (Hypki & Giersz 2013; Giersz et al. 2013), however, are computationally cheaper and useful for large grid explorations, but relies on more stringent assumptions (such as spherical symmetry). Age and metallicity estimates are usually obtained through isochrone fitting of color-magnitude diagrams (CMDs; Rosenberg et al. 1999; Zoccali et al. 2003; Dotter et al. 2010; Leaman et al. 2013; Valcin et al. 2025, 2026), a method that is relatively robust considering the overall difficulty in determining ages, but that is not easily scalable to the volume of data expected from next-generation surveys.

In this context, machine learning, and in particular deep learning (DL), offers promising alternatives (Bialopetravičius & Narbutis 2020a,b; Chardin & Bianchini 2021; Hiegel et al. 2023; Guiglion et al. 2024). DL approaches have already shown great potential in automating the classification of extragalactic star clusters (Zhang et al. 2025), estimating their ages (Boin et al. 2026; Viaña et al. 2026) and dynamical properties (Pasquato & Chung 2016; Askar et al. 2019; Bialopetravičius et al. 2019; Pasquato et al. 2024), and more recently, estimating global parameters of GCs directly from imaging data (Chardin & Bianchini 2021). The latter introduced the proof-of-concept algorithm π -DOC (Predicting Images for the Dynamics Of globular Clusters), trained on mock photometric observations generated from forward-modeled N -body simulations (Miholics et al. 2016; Bianchini et al. 2017). The study demonstrated that deep networks can infer underlying physical quantities of GCs, such

as their internal mass distribution, age, and distance, directly from photometric images. This approach offers not only an efficient and scalable method, but is also one of the first tools capable of mapping a two-dimensional mass distribution without assuming a mass-to-light (M/L) ratio.

The initial proof-of-concept algorithm relied on idealised mock images constructed from low-mass GC simulations, lacking realistic observational uncertainties. Neither instrumental noise nor galactic field-star contaminants were modeled, and consequently the network cannot reliably be employed on observational images unless the GC is isolated beforehand. Moreover, the limited parameter coverage restricted applicability to just five Galactic GCs. Such limitations reflect a synthetic gap: a mismatch between idealised training data and complex real-world observations that typically degrades performance. Nevertheless, these results demonstrated the strong potential of DL approaches for large-scale analyses of GCs.

In this study, we address the limitations of the previous proof-of-concept version of π -DOC. In particular, we extend the underlying data with more massive GC simulations from both the ROLLIN' suite (Bianchini et al. 2026; Marklund et al. 2026) and the MOCCA simulations (Askar et al. 2025; Zhao et al. 2026). In addition, we combine our synthetic mock images with real observations of field stars, simultaneously modelling instrumental and physical uncertainties from observations. These steps ensure we narrow the synthetic gap. The underlying data and the construction of realistic mock images are described in Section 2.

The previous version of π -DOC was split into two parts: an autoencoder that mapped the input image to a two-dimensional mass distribution, and a classical convolutional classifier that predicted the cluster's age and distance from the same input. This separation is inefficient, since both parts must be trained independently, and precludes a joint latent representation — which could otherwise be useful given that certain outputs, such as age and the mass distribution, are inherently related (Baumgardt 2017; Bianchini et al. 2017). For the new architecture we have combined the two networks into one multi-task network, which now utilises multi-band photometric images, and in addition predicts a decontaminated version of the observed cluster (i.e. removal of field stars), as well as the cluster's ellipticity and corresponding position angle. A full description of the new architecture can be found in Section 3, and the code is available online¹.

As a concrete test case for developing an efficient algorithm intended for large-scale surveys, we apply π -DOC to GCs in the Andromeda galaxy (M31) using archival HST data. In this work, the network is restricted to GCs that are at least partially resolved. This constitutes an initial step toward a more general tool for extragalactic GC systems.

The remainder of the paper is structured as follows. In Section 4 we assess the performance of π -DOC using realistic mock observations, exploring biases and its robustness to noise. In Section 5 we assess the reliability of estimates on real observations as a function of their representational similarity to the training data. In Section 6, we apply our algorithm to M31 GCs, present the main results, and compare π -DOC's estimates with the literature. We conclude with a brief discussion in Section 7 and conclusions in Section 8.

2. Real and synthetic data

π -DOC aims to automate the detailed analysis of GCs directly from multi-band photometric observations. As a supervised neu-

¹ <https://github.com/arnmarklund/pidoc>

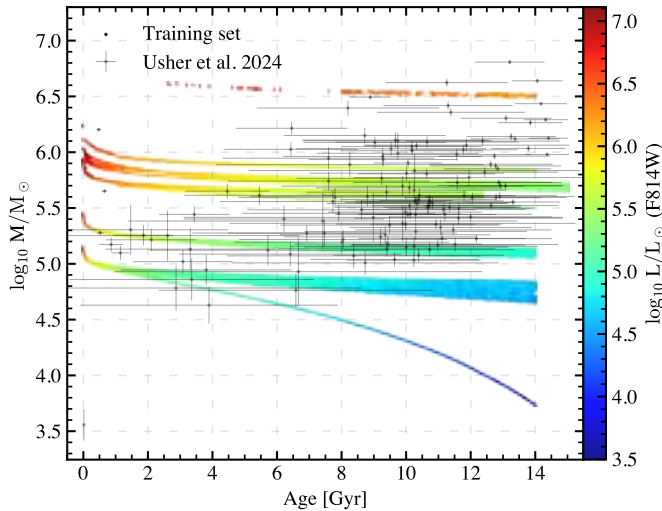


Fig. 1. Comparison of the mass–age relation for our mock images and the M31 globular clusters from (Usher et al. 2024) located within the PHAT and PHAST footprints. The mock images are colour-coded by their total luminosity in the F814W passband, while the observed GCs are shown as black points. Our mock sample broadly covers the parameter space of the M31 GC system, and, given the age uncertainties of the observed clusters, the most important aspect is a representative sampling of the mass distribution. Nonetheless, some gaps remain, particularly at the highest masses.

ral network, its training data must be both physically relevant and representative of real observations. We first describe the real observational data used (Section 2.1), then explain how these data inform the construction of realistic synthetic mock images (Sections 2.2 and 2.3), reducing the synthetic gap. We summarise the datasets used throughout the paper in Section 2.4.

2.1. Real GC data

In this work we consider GCs in M31, using archived HST data from the Panchromatic Hubble Andromeda Treasury (PHAT) survey² (Dalcanton et al. 2012; Williams et al. 2014, 2023), as well as the recent extension to the southern parts of M31 (PHAST, Chen et al. 2025). The PHAT and PHAST surveys each cover about a third of M31’s star forming disk, divided into 23 and 13 bricks, respectively. The former is observed in six passbands: F110W, F160W, F275W, F336W, F475W and F814W, whereas the PHAST survey only has the latter two available for mosaic images. We consider all bricks, but due to low signal-to-noise (S/N) we choose to only work with the F336W (where applicable), F475W and F814W passbands (Williams et al. 2023). We use drizzled images (multiple exposures), and align them north to east, conserving the flux and assuming a common scale of 0.04 arcsec/pixel using the `reproject` package³. We select GCs from the Peacock et al. (2010a,b) catalogue that lie within the boundaries of the PHAT and PHAST footprints, amounting to a total of 349 GCs. Every image of a GC we consider has a $12'' \times 12''$ field of view (FoV), which is large enough to include typical GCs in M31 to a few times the half-light radii (typical half-light radii of M31 GCs are $\sim 1\text{--}2''$ Barmby & Huchra 2001; Huxor et al. 2014) at a distance of ~ 785 kpc (McConnachie et al. 2005).

In addition, we sample every brick and keep a total of 12264 images from the PHAT survey that are free of GCs and other star clusters, with the intention of using these as backgrounds⁴ to our synthetic dataset (see Section 2.3 for details).

2.2. Simulated data

Given the details of the real observational images, it is important that our synthetic data, which will be used to train π -DOC, reflect the same underlying features and parameter space present in the observations. In other words, we must ensure that the synthetic data share both the underlying physical properties and the observational characteristics of real images. We address the first concern by considering two different suites from state-of-the-art simulations:

1. The first and main one, refers to the ROLLIN’ suite (Bianchini et al. 2026; Marklund et al. 2026), and consists of initially rotating, axisymmetric direct N -body simulations with an initial amount of 250k–1.5M stars with a metallicity of $[\text{Fe}/\text{H}] = -1.3$ dex, including prescriptions for stellar evolution and an external tidal field. These initially rotating models lead to a wide range of GC-morphologies due to internal rotation and interactions with the external tidal field (Marklund et al. 2026), but remain limited in total mass and density due to the significant computational cost of direct N -body integrations. As such, they particularly resemble low-density GCs in M31 and the Milky Way. In total we use ten such simulations with an initial amount of 250k stars, three with 500k stars, and one with 1.5M stars.
2. The second group refers to MOCCA simulations, and include four models of 47 Tucanae ($[\text{Fe}/\text{H}] = -0.78$ dex), one model for ω Centauri ($[\text{Fe}/\text{H}] = -1.5$ dex), and five additional high-density models ($[\text{Fe}/\text{H}] = -1.6$ dex) which are described in more detail in (Askar et al. 2025; Zhao et al. 2026, with two models from private communication Askar 2026). MOCCA simulations relax the computational constraints in the direct N -body method by assuming spherical symmetry. As such, while they do not account for a morphological variety, they allow us to run simulations with considerably more stars, higher total masses, as well as higher densities. In total we use one such model with an initial amount of 1M stars, two with 1.5M stars, five with 2M stars, one with 2.4M stars, and one with 12M stars (roughly resembling ω Centauri). These models correspond to high-density and massive M31 and Galactic GCs.

These two groups of simulations therefore complement each other and ensure that the parameter space of our full dataset broadly matches that of real GCs in M31. Figure 1 shows the masses and ages of the mock images in our dataset (coloured points), together with the corresponding mass and age estimates for GCs in M31 from Usher et al. (2024). Although a few gaps remain, the dataset overall covers most of the relevant parameter space.

The underlying data to be used for each mock image corresponds to a snapshot from one of these simulations at different time steps. For the ROLLIN’ simulations, the frequency of the snapshots vary between 1 – 15 Myr depending on the simulation, whereas for the MOCCA simulations it is either 50 or 100 Myr, depending on the simulation and the age of the GC. As such,

² <http://dx.doi.org/10.17909/T91S30>

³ <https://reproject.readthedocs.io/>

⁴ Note that we call these images backgrounds, but in principle they could actually be foreground images.

different simulations will be more or less prominent in the overall dataset, and in particular, there will be a bias towards the ROLLIN' simulations. This is also partly by construction as we are interested in exploring the morphology of GCs.

However, to mitigate biases within the dataset, we restrict the number of unique snapshots of simulations with 250k stars to $\min(N, 2000)$, while for the more massive ROLLIN' simulations (500k-1.5M stars initially) the total number of snapshots are set to $\min(N, 3000)$ where N is the total amount of available snapshots from a simulation. This leaves us with 33k unique ROLLIN' snapshots. On the contrary, we consider all available snapshots from the MOCCA simulations. In total, we have $\sim 40k$ unique snapshots from a total of 24 simulations.

To generate mock images from these simulated models of GCs that properly treat typical observational characteristics of HST M31 data, we have developed a forward modelling algorithm, CREMANT (Creating REalistic Mock imAges for Numerical simulaTions)⁵. The algorithm makes use of

- (i) the flexible stellar population synthesis (FSPS) code (Conroy et al. 2009; Conroy & Gunn 2010) to compute individual stars' magnitudes in the passbands we consider for our mock images. The stellar parameters from the simulations that are passed to FSPS are mass, radius, luminosity, and metallicity, and the effective temperature and surface gravity are calculated in FSPS. These stellar properties (excluding metallicity which is passed as an input to the simulations) are, in turn, provided by the stellar and binary evolution from the SSE and BSE prescriptions of Hurley et al. (2000, 2002) incorporated within the star cluster evolution codes (we refer to Kamlah et al. 2022, for the stellar evolution implementation in our N -body simulations). Within FSPS, we use PADOVA isochrones (Bressan et al. 2012) and the BASEL spectral library (Lejeune et al. 1997, 1998; Westera et al. 2002).
- (ii) two angles and a distance, as free parameters, that define geometrical projections. The field of view of our images ($12'' \times 12''$) and the pixelscale ($0.04''$) are also free parameters⁶ in the algorithm, and together define a two-dimensional pixel-grid onto which the particles from a simulation can be projected. To generate a projection, we first define a versor, which represents a rotation axis with respect to the coordinate axis of the simulation, and a rotation angle $\phi \in [0, \pi)$. Together, the versor and rotation angle determine the inclination⁷ of the cluster. The projected cluster can subsequently be rotated in the image plane by an additional angle $\theta \in [0, 2\pi)$, applied around the axis perpendicular to the projection plane. These two rotations fully specify the apparent orientation of the simulated cluster. The cluster is then placed at a chosen distance which, together with the apparent orientation, determines the relative positions for each star within the pixel-grid.
- (iii) point-spread functions (PSFs), derived from non-saturated bright stars in the outer regions of M31 (PHAT brick 22 with

each passband aligned north-east) using DrizzlePac⁸. We include four distinct PSFs per passband, including STScI's empirical model⁹, all accessible via CREMANT. Multiple PSFs are included to prevent the network from memorising an imperfect PSF model with respect to the backgrounds' PSF. We oversample each mock image and PSF with a factor of four to allow for sub-pixel variations before convolving the images to produce the final mock image.

2.3. Modelling noise

Our mock cluster images are idealised source-only realisations, effectively corresponding to a noise-free limit in which the GC flux is known without photon-counting noise and with no background. To generate realistic observations, and to thereby reduce the synthetic gap, we combine our mock images with backgrounds from real observations following the steps outlined in Bialopetravičius et al. (2019). We first transform the mock images into units of counts per second, matching the background images, and then:

1. compute the median pixel value of the background and add it to each pixel of a mock cluster image;
2. draw a Poisson realisation for each pixel, in total counts, with the mean set to the corresponding pixel value;
3. subtract the median offset and add the real background image;
4. convert the pixel values to magnitudes (ST system).

This approach ensures that we model both the instrumental and background noise associated with an observation. Because the mock images represent noise-free source models, we temporarily shift them by the median background level of a real field before applying Poisson sampling to the total count rate. The temporary offset places the source model on a realistic count scale before Poisson sampling. We then remove the offset and add the full real background image, so that the final mock preserves both the observed background structure and realistic counting statistics.

We do not apply any magnitude cuts, since the background already sets the effective baseline. During training (see Section 3.4), 10% of the mock images are also generated background-free to improve generalisation to real HST data with negligible contamination¹⁰. Because CREMANT currently does not implement extinction, each background is corrected for Milky Way extinction using dustmaps¹¹ (Schlegel et al. 1998; Green 2018).

This procedure is carried out on the fly during training, effectively extending our total dataset by four orders of magnitude, since each mock image can be paired with any of the 12264 backgrounds. A qualitative comparison between examples of our mock images and real GCs in M31 is shown in Fig. A.1.

2.4. Building the datasets

For each snapshot of a simulation, we create a total of 6 mock images, with three different projection angles, and each angle at two different distances. In total, we create approximately

⁵ <https://github.com/arnmarklund/cremant>

⁶ The pixel scale is set by the choice of telescope (HST), and the field of view is chosen to minimise image size while including as much of the GC as possible. In particular, we used the more massive, low-density and extended ROLLIN' models at the distance of M31 to set a reasonable FoV, and then cross-checked against observed M31 GCs, bearing in mind that their full spatial extent is not known exactly.

⁷ We consider the inclination with respect to the z-axis and the line of sight. The z-axis corresponds to the rotation axis for the ROLLIN' simulations.

⁸ https://github.com/spacetelescope/hst_notebooks/tree/main/notebooks/DrizzlePac

⁹ <https://www.stsci.edu/hst/instrumentation/wfc3/data-analysis/psf>

¹⁰ In such cases, we omit step 3 above, and stars fainter than the typical HST background therefore remain visible.

¹¹ <https://dustmaps.readthedocs.io/en/latest/index.html>

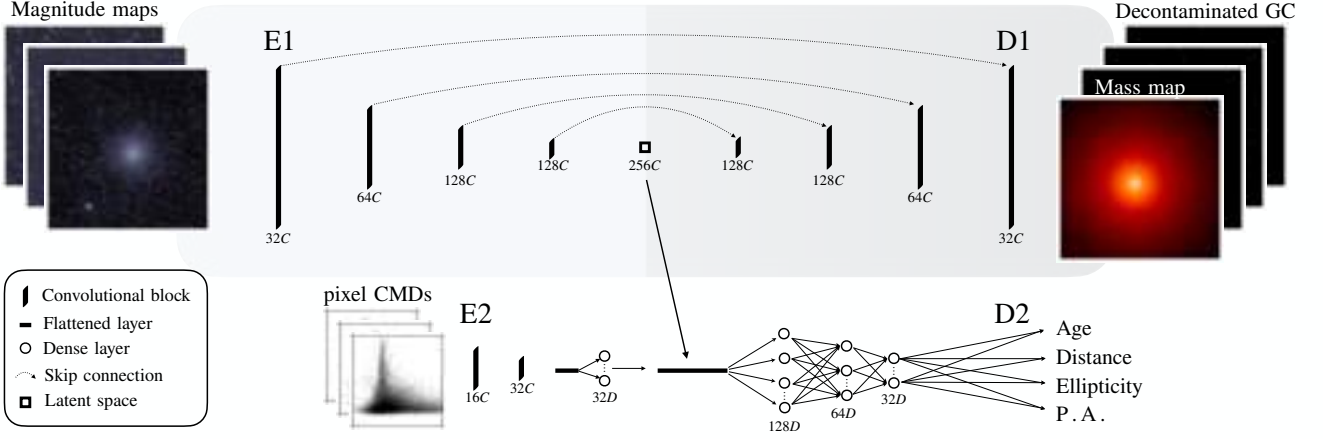


Fig. 2. Architecture of π -DOC. The different encoders are labeled E1-E2 and the respective decoders are labeled D1-D2. The corresponding number of filters (channels) are indicated below the blocks. Skip connections are indicated by dotted arrows. The regression head (D2) combines the latent features from each of the encoders, and the number of dense layers are indicated beneath the symbols.

210k mock images of GCs from the ROLLIN’ simulations, and roughly 10k images from the MOCCA simulations.

The distributions of parameters for the dataset are chosen such that they cover the expected ranges of parameters for GCs in M31, where possible. [Holland \(1998\)](#) put most of the GCs at similar distances to M31 itself, and from the Galactic population of GCs we know they may extend as far as ≥ 100 kpc from the Galactic Center (see e.g. [Harris 2010](#)). As such we choose distances to be sampled from a uniform distribution centred at the distance of M31, 785 ± 100 kpc (here using the distance estimate to M31 from [McConnachie et al. 2005](#)).

Some quantities are set directly by the simulations, such as the age (sampled uniformly in linear scale¹²), total luminosity, and total mass (see [Bianchini et al. 2026](#); [Marklund et al. 2026](#)). The projected images, the two-dimensional mass distribution, ellipticity, and the corresponding position angle (P.A.) for the minor axis¹³, are also set by the simulations, but further depend on the two geometrical projection angles defined in CREMANT (Section 2.2), as well as the distance. These angles and the versor are all sampled uniformly.

We compute a GC’s apparent ellipticity $e = 1 - b/a$ using the second-moment tensor method on the stellar distribution within the FoV, where a and b are the major and minor axes, respectively. This robust approach reliably traces the cluster’s intrinsic morphology ([Fréour et al. 2026](#); [Marklund et al. 2026](#)), facilitating clearer interpretation of our results. The position angle is measured east of north over $[0^\circ, 180^\circ)$.

Apart from the observational images, we also construct pixel colour-magnitude diagrams (CMDs) that are 64×64 pixels¹⁴. In cases where three (two) passbands are used, we get three (one) possible pixel CMDs: F475W vs F336W–F475W, F814W vs F336W–F814W, and F814W vs F475W–F814W.

We split the entire datasets into three parts: training ($N_{\text{train}} \sim 130\text{k}$), validation ($N_{\text{val}} \sim 44\text{k}$), and testing ($N_{\text{test}} \sim 50\text{k}$). The three sets are constructed in such a way that no mock image exists in more than one of the sets. This also applies to the real backgrounds, which in turn are distributed among the three sets in proportion to that set’s number of mock images (e.g., roughly 60% of the backgrounds are in the training dataset).

In order to get a diverse training dataset, it is customary to perform data augmentation. We combine the mock images and the real backgrounds on-the-fly when loading each batch, effectively obtaining a new set of images for every epoch. Each image is also rotated either 0° , 45° , 90° , or 135° .

3. Network

In this section we describe the new architecture of π -DOC as well as the corresponding inputs and outputs. π -DOC is a convolutional autoencoder based on the U-net architecture ([Ronneberger et al. 2015](#)), but it has been deeply modified to handle hybrid-inputs and to produce multi-modal outputs. In particular, it is now capable of handling multi-band photometric images and pixel CMDs. Inputs are described in more detail in Section 3.1. We note that the inclusion of pixel CMDs was found to significantly improve age and distance estimates in early testing of the new network architecture, in particular due to a more explicit encoding of the shape of the associated isochrones. There are also three completely new outputs: a decontamination of field stars from the input multi-band photometric image, and the GCs’ ellipticity and corresponding position angle for the minor axis. We describe all outputs in more detail in Section 3.2.

The additional inputs and outputs have also resulted in a re-work of π -DOC’s overall design, and we describe the corresponding architecture in more detail in Section 3.3. Additionally,

¹⁴ We note that using pixel CMDs with an increased resolution (128×128 , 256×256) did not improve performance.

¹² The distribution is only approximately uniform, as the largest N -body simulation was evolved only up to 12 Gyr, three MOCCA simulations were evolved up to 15 Gyr, the ω Cen equivalent model is additionally under-sampled at all evolutionary stages due to the computational cost of its magnitude calculations, and early snapshots are entirely absent, while all the remaining simulations were run up to 14 Gyr.

¹³ In projection, the minor-axis position angle is equivalent to the major-axis angle shifted by 90° . We adopt the minor axis here as a matter of convention and for consistency with prior studies (e.g. [Fréour et al. 2026](#); [Marklund et al. 2026](#)).

because the PHAST survey is available only in two passbands, the network is available for both two and three passbands, and we denote them π -DOC₂ and π -DOC₃, respectively.

3.1. Input data

The main inputs to the network consist of the multi-band photometric images alongside pixel CMDs. π -DOC₂ uses two magnitude maps and one CMD, and π -DOC₃ uses three magnitude maps and three CMDs. We also experimented by explicitly providing colour maps as part of the inputs, but did not find this to yield any improvements. This is reassuring because it suggests that such information is redundant and already encoded in the current inputs. In contrast to the proof-of-concept version of π -DOC (Chardin & Bianchini 2021), the input images are not smoothed. See Section 2 for more details on the construction of the input data.

All inputs are standardised, though not uniformly. The magnitude maps across all passbands are standardised jointly — using a single combined mean subtracted from every band — rather than independently per band, and without enforcing a unit standard deviation. This preserves the relationship between magnitudes and colours, so that even after standardisation, the difference between two bands still corresponds directly to their colour. The pixel-wise CMD number counts, by contrast, are standardised to zero mean and unit standard deviation. All standardisation constants are computed once by iterating over the training data (e.g. one epoch, with each mock image combined with a background; see Section 2.3). Since noise augmentation is applied on-the-fly, no epoch sees exactly the same images; however, given the size of the training set (~ 130 k images), the standardisation constants computed from a single epoch are statistically stable and not meaningfully affected by this variation.

3.2. Output data

π -DOC is tasked to decontaminate the input images in all passbands from the background, map the corresponding two-dimensional mass distribution within the FoV (same pixel-scale as the input images), and predict the cluster’s age, distance, ellipticity, and position angle for the minor axis (P.A.). Instead of predicting P.A. directly, the network predicts $\cos(2\text{P.A.})$, $\sin(2\text{P.A.})$; the resulting P.A. is then calculated using the two-argument arctangent (atan2) as $\frac{1}{2} \text{atan2}(\sin(2\text{P.A.}), \cos(2\text{P.A.}))$. This approach removes large errors at the periodic boundary where 0° and 180° are degenerate.

We also apply a Gaussian smoothing kernel to the mass maps ($\sigma = 4$). This is to avoid the network having to deal with predicting highly pixelated images, especially in the outskirts of the clusters where the stellar counts are sparse. Note that we have tested applying such a smoothing also on the input and output magnitude maps, but this did not improve performance. The input images and the decontaminated output images therefore remain consistent with HST’s pixelscale.

All of these outputs are standardised during training except for the position angle since we infer $\sin(2\text{P.A.})$, $\cos(2\text{P.A.})$. The decontaminated magnitude maps are standardised to be consistent with the input magnitude images (see Section 3.1), and the remaining outputs are standardised to have a mean of zero and a standard deviation of one.

3.3. Network architecture

The inclusion of additional input data, such as the CMDs, enables more complex operations within the network, encouraging it to extract both complementary and distinct features from multiple representations of the same underlying system. The new architecture reflects this, and we show a schematic overview in Fig. 2. There are four different components, which can be split into two encoding parts (E1 and E2) and two decoding parts (D1 and D2)¹⁵:

- E1 is an encoder applied on input magnitude maps (first row of Fig. 2) that consists of five convolutional blocks.
- E2 is a feature extractor applied on CMDs (second row of Fig. 2). The convolutional blocks are slightly different from those employed in E1.
- D1 is a decoder (first row of Fig. 2) consisting of five deconvolutional blocks and decontaminates the input magnitude maps.
- D2 is the regression head in the second row that predicts the scalar outputs (age, distance, ellipticity, and position angle) from the concatenated and flattened latent spaces of the two encoders E1 and E2.

Finally, to reduce overfitting we use dropout and set it to 0.1 for each version.

3.4. Training

For all versions of π -DOC, the training is governed by the same loss function, which we take to be the weighted sum of each output’s mean-square-error (MSE). Note that we also experimented with incorporating a perceptual loss function for predicting the mass and luminosity, as per Chardin & Bianchini (2021), but we did not observe it to yield any improvement. As a result, we consider the total loss function as

$$\mathcal{L} = \sum_{i=1}^I w_i \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|^2, \quad (1)$$

where I denotes the number of outputs, each $\mathbf{y}_i$ denotes one output: the decontaminated magnitude maps which is considered as a singular output regardless of the number of passbands, mass map, age, distance, ellipticity, and position angle. Consequently, $\hat{\mathbf{y}}_i$ is its estimate by the network. w_i denotes a scaling factor that is set to ensure each output MSE is in the same range¹⁶.

Both networks have been trained on 4 NVIDIA H200 GPUs. We use the ADAM optimizer (Kingma & Ba 2014), and a learning rate of $\eta = 10^{-4}$. Moreover, we implemented a learning-rate scheduler halving after 2 epochs without validation improvement and an early stopping after 5 epochs. Fig. 3 shows the training and validation performance over epochs for each network. Training ran for a maximum of 50 epochs to save computational time with no early stopping triggered, yielding near-converged models with stable validation performance (total training required > 1000 GPU hours for each model).

¹⁵ The latter of these parts illustrate a feature extractor and a regression head rather than an encoder and decoder, but for simplicity we denote them as E2 and D2.

¹⁶ The weights are set to one for all outputs except for the magnitude and $\cos(2\text{P.A.})$, $\sin(2\text{P.A.})$, which are set to 0.2, 0.5, and 0.5, respectively. Furthermore, for GCs with $e = 0$ we dynamically set a zero-weight associated to the P.A. since it is not defined.

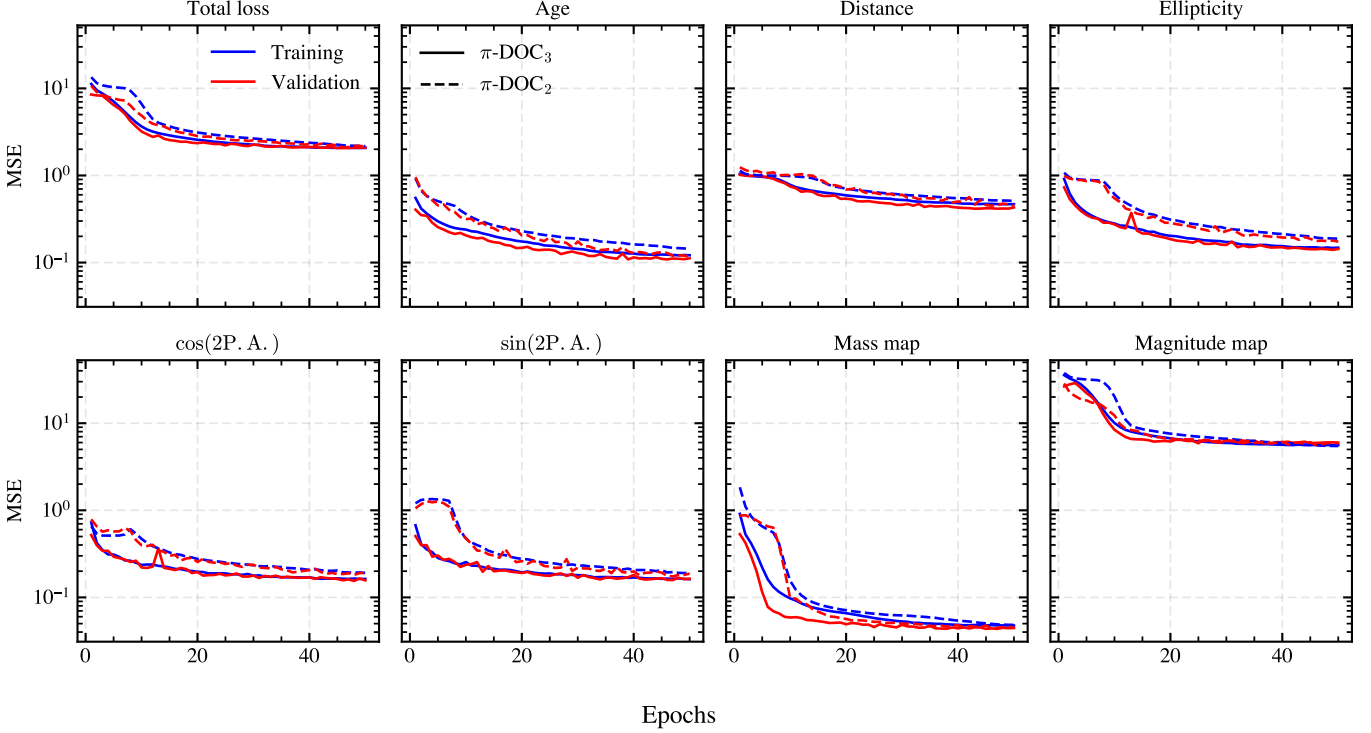


Fig. 3. Training curves for the four π -DOC variants considered in this study. Blue (red) lines correspond to the training (validation) MSE per output across epochs. The three-passband networks converge faster and achieve lower overall MSE than their two-passband counterparts. The addition of colour maps and an adapted architecture with two encoders and shared skip connections mainly improves the accuracy of the predicted mass maps. We interpret this as a result of the larger (but not deeper) network capacity. The vertical black dotted line marks epoch 50, which is used to compare the networks' performance.

Table 1. MSE of the two networks' validation performance for the different outputs.

MSE	π -DOC ₂	π -DOC ₃
Magnitude map	5.86	6.00
Mass map	0.05	0.04
Age	0.12	0.11
Distance	0.44	0.43
Ellipticity	0.17	0.14
Position angle	0.19	0.16
Total weighted	2.17	2.09
Epoch	50	50

Notes. MSE for each quantity. The MSE for the P.A. is taken to be the average between $\sin(2P.A.)$ and $\cos(2P.A.)$.

We indicate the total MSE, and the output-specific MSEs, for the validation dataset for each version of π -DOC in Table 1. We find that the inclusion of F336W marginally improves performance for all outputs except decontamination. We attribute this to two competing effects: the additional passband provides more constraining information, but also increases the dimensionality of the decontamination task, which must isolate the GC signal from the field and background consistently across all bands. For decontamination, this added complexity appears to outweigh the benefit of the extra information.

4. π -DOC's performance on mock images

For the assessment of the network's performance on mock observations, we begin by visually inspecting some of the predictions. Throughout this section, all predictions are obtained with π -DOC₃ unless explicitly mentioned otherwise. Fig. 4 shows a mosaic of four selected mock images from the test set, using the three passbands to construct a pseudo-RGB image. These mock images were chosen to be representative of different cases we expect to find: (a) a relatively young and low-density cluster with low background contamination (high S/N and a ROLLIN' model), (b) an old but fairly dense (bright and massive) cluster in a high background contamination (low S/N and a MOCCA model), (c) an older fainter cluster in a higher background contamination (low S/N and a ROLLIN' model), and (d) an old dense cluster in a low background contamination (high S/N and a MOCCA model). The second and third row show the decontamination part of the network¹⁷, and also includes the scalar regression output for the distance and age, as well as the total luminosity and half-light radius, which are derived from the decontaminated map.

Across the four examples, the decontamination and associated scalar outputs remain close to the ground truth. In the high S/N examples individual stars are retained, though their colour is not guaranteed to be correctly inferred. Instead, there is a tendency to assign similar magnitudes per pixel across the various passbands with the resulting GC appearing whiter. The success-

¹⁷ Pixels in the visualisation of these images are confined to magnitudes between 25.7 and 31. Note, however, that π -DOC's predictions contain pixels below and above these values.

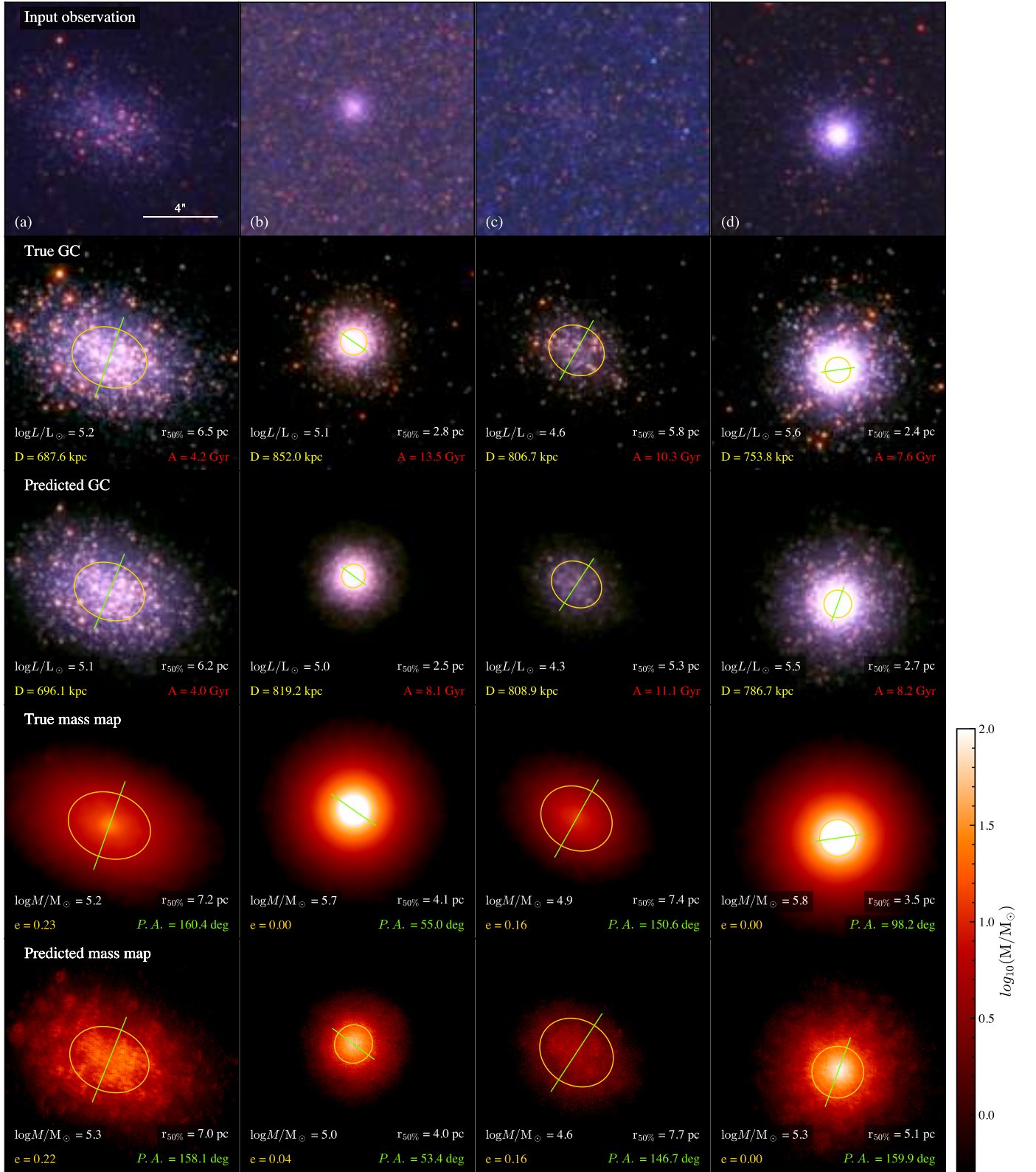


Fig. 4. Mosaic example of four different mock images (top row), the true decontaminated cluster (second row), the predicted decontaminated cluster (third row), the true mass distribution (fourth row), and the predicted mass distribution (bottom row). The decontaminated panels also indicate the true and predicted values for total luminosity and half-light radius within the FoV, distance, and age. Similarly, the mass distribution panels indicate the total mass and half-mass radius within the FoV, the ellipticity, and the position angle. The golden ellipse in each panels indicate the ellipticity and has its semi-major axis placed at the half-light or half-mass radius, whereas the green line indicates the position angle of the minor axis. The true and predicted decontaminated cluster images are scaled differently to the input observations in order to highlight the differences in decontamination across the full extent of the cluster.

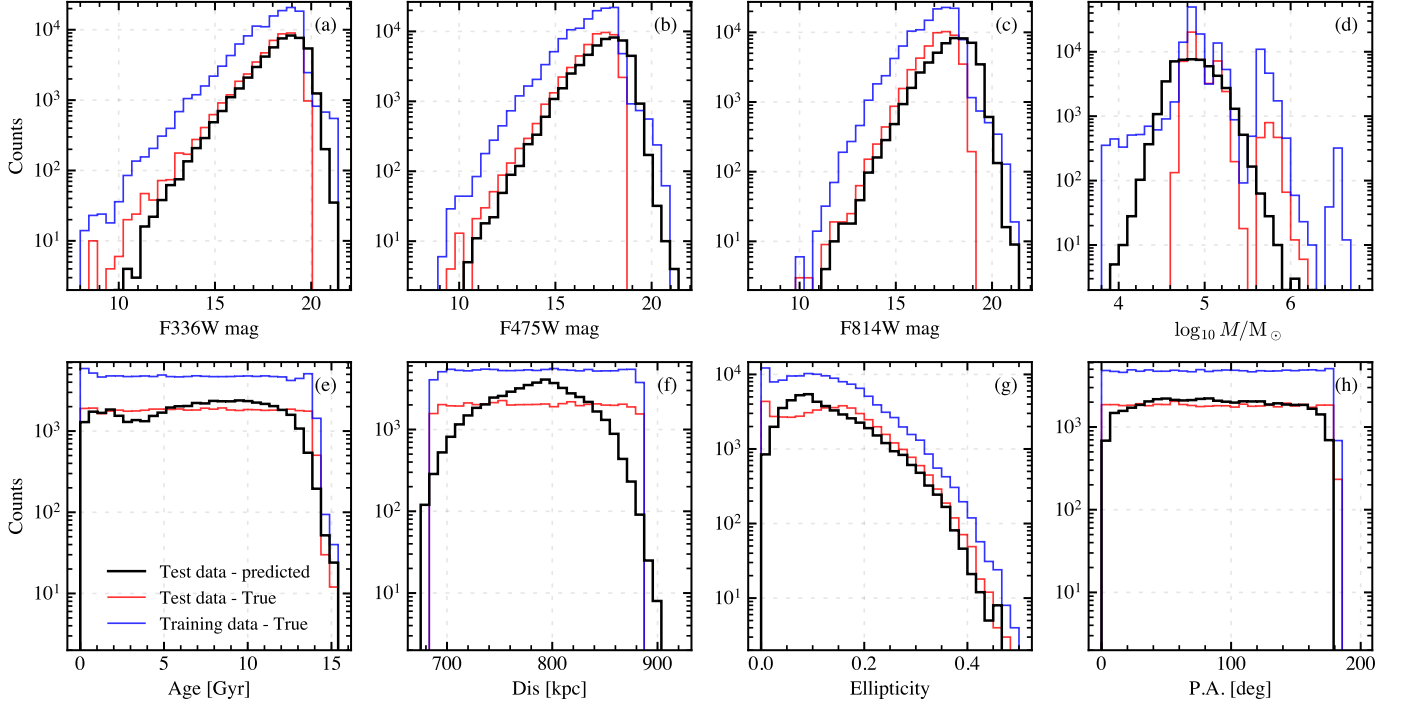


Fig. 5. Distribution of the predictions for the main outputs of π -DOC (black lines), including the corresponding distributions for the test data (red lines; ground truths to predictions) and the training data (blue lines). Top row: (a) F336W magnitudes, (b) F475W magnitudes, (c) F814W magnitudes, (d) total mass in log-scale. Bottom row: (e) age, (f) distance, (g) ellipticity and, (h) position angle.

ful decontamination of the clusters in the brighter backgrounds, panels (b) and (c), demonstrates that π -DOC can distinguish the overall diffuse distribution of the GC from the background. On the other hand, not all stars (or pixels) are successfully recovered, where, in particular, there is an emphasis on stars further from the centre of the GC. This, however, does not appear to affect global quantities significantly.

The remaining two rows show the true and predicted 2D mass distributions, and the corresponding total mass and half-mass radius (derived directly from the mass maps). The total mass is constrained to within a factor of two for the two ROLLIN' models in panels (a) and (c), while the MOCCA models in (b) and (d) amount to an underestimation by a factor of ~ 5 and ~ 3 , respectively. This is an indication that the network struggles to generalise between our ROLLIN' and MOCCA models.

Our training data are dominated by N -body models, so high-mass clusters are primarily represented by the comparatively small number of MOCCA models included in the training set. These results may therefore partly reflect π -DOC's limited ability to generalise in this poorly sampled regime, rather than a genuine astrophysical trend. An alternative, and not mutually exclusive, explanation is the retention and spatial distribution of non-visible mass within the GCs (e.g. stellar remnants). Example (d) hosts both an intermediate-mass black hole ($M \sim 6000 M_\odot$) and a substantial white dwarf (WD) population ($M \sim 1.3 \times 10^5 M_\odot$), together accounting for roughly 30% of the missing mass, while the non-visible remnant population in example (b) accounts for 25% ($M \sim 1 \times 10^5 M_\odot$) — implying remnant mass fractions of order 15-25% in the MOCCA models. We find a similar fraction for the ROLLIN' models, but their spatial distributions differ substantially: in the N -body examples, WDs closely trace the overall mass distribution (the ratio of Lagrangian radii are $r_{10,WD}/r_{10,all} \sim 1$, $r_{30,WD}/r_{30,all} \sim 0.9$, and $r_{50,WD}/r_{50,all} \sim$

$0.84-0.88$)¹⁸, whereas in the MOCCA examples the WDs are considerably more centrally concentrated ($r_{10,WD}/r_{10,all} \sim 0.8$, $r_{30,WD}/r_{30,all} \sim 0.7$, and $r_{50,WD}/r_{50,all} \sim 0.65 - 0.74$).

We suggest this may contribute to π -DOC's underperformance on MOCCA-like clusters. Since the network infers total mass from the visible light distribution alone, it must implicitly learn a visible-to-total mass (or alternatively a spatially dependent M/L ratio) mapping calibrated mainly on the N -body-dominated training set, where remnants are not strongly concentrated relative to the total mass. Applied to MOCCA-like clusters, where a comparable remnant fraction is instead concentrated within a much smaller region, this mapping would be expected to undercorrect the innermost pixels, while the greater compactness of MOCCA clusters leaves less room to compensate. We explore these possibilities in more detail in Section 7.

The ellipticity of the two denser, more massive (brighter) clusters in columns (b) and (d) is zero, since MOCCA simulations assume spherical symmetry. In panel (b), π -DOC infers a small but non-zero ellipticity of 0.04. This offset may provide a rough sense of the absolute precision achievable by the model in a low S/N regime. In panel (d), the ellipticity is correctly recovered. Although the position angle in panel (b) is constrained to within 2° it is merely a coincidence as neither P.A. in panels (b) and (d) are defined for $e = 0$ (we only include the P.A. for completeness). By contrast, both the ellipticity and position angle are recovered for the two ROLLIN' models.

Although these four mock images represent only a small subset of our full test set, they illustrate how π -DOC performs under realistic observational conditions. The network successfully decontaminates even faint clusters in bright crowded fields, though limitations appear in this low S/N regime: example (b),

¹⁸ These are the radii within which 10%, 30%, and 50% of the mass is confined.

which has high contamination, shows the largest age discrepancy (~ 5 Gyr vs $\lesssim 0.7$ Gyr for the other cases) as well as a significant mass offset (also true for example c) and the largest distance error (~ 30 kpc, similar to the other MOCCA model in panel d). For the two ROLLIN' models, distances are consistent to within ~ 10 kpc. Overall, π -DOC reliably infers GC properties when the S/N is sufficiently high.

4.1. Performance on test set

Qualitatively, the performance shown in Fig. 4 is representative of the full test sample. Fig. 5 shows the prediction distributions for all π -DOC outputs on the test set, alongside the corresponding training and test parameter distributions. The training data span the full parameter space of the test set, and the predicted distributions closely track the test distributions across this range, indicating the broad applicability of π -DOC.

Panels (a-c) show decontamination performance in the three photometric passbands. The integrated magnitudes are slightly overestimated across the entire range (i.e. their brightness is underestimated), and this trend is more severe for the faintest clusters (see the tail at large magnitudes), which is expected since faint clusters are harder to distinguish from the background.

The total mass distribution in panel (d) shows two distinct peaks in the test set, while the network instead predicts a more unimodal distribution and underestimates both the least massive clusters and the most massive ones. This largely follows from the decontamination results above: clusters whose brightness is underestimated typically also have their total mass underestimated. At the massive end, this may also reflect a bias toward lower mass predictions, likely affected by the relatively poor representation of massive simulations in the training data noted above.

The training and test distributions are uniform for both age (panel e) and distance (panel f). The predicted age distribution is also fairly uniform, aside from two modest peaks near 1–2 Gyr and 9 Gyr. Distance predictions, by contrast, cluster toward the mean of the test distribution. This likely reflects the intrinsic difficulty of constraining distance, but also mirrors a well-known behaviour in neural networks (Kang et al. 2024), which tend to perform less reliably near the edges of the training domain: because the network implicitly learns the bounds of the parameter space, it avoids extrapolating beyond them and instead biases predictions toward interior values.

Predictions of ellipticity (panel g) and position angle (panel h) closely match the ground truth across the full range. Small ellipticities ($\lesssim 0.1$) are overestimated, likely reflecting a bias against predicting exactly zero for the MOCCA models (see e.g. Fig. 4). At these small ellipticities, the position angle is either undefined ($e = 0$) or poorly defined, since there is no clear major or minor axis; large position-angle errors are therefore expected in this regime and are not to be considered a concern.

Overall, π -DOC₂ and π -DOC₃ agree well across all predicted quantities, with the former showing larger absolute uncertainties. For an explicit comparison between the two models, see Appendix B.

4.2. Biases within the network

The previous section explored how the network retains the overall distributions of the test set. Here, we explicitly aim to address the cases where the predictions are most uncertain. Fig. 6 shows a direct comparison of the differences between the predictions and the ground truth of the test data, as well as the correspond-

ing correlations between the quantities. We refer to the difference between a predicted quantity and its ground truth as the error. To facilitate the discussion below we keep the ground truth magnitude for the F336W passband in column (a) rather than the corresponding error. The diagonal further indicates the mean and 1σ of each distribution.

Column (a) shows that the errors increase for most quantities the fainter the clusters are. As previously mentioned, this likely reflects an increased difficulty in distinguishing the GC from the background. Indeed, the majority of the GCs in our data set are relatively faint (see also Fig. 5), having an average integrated magnitude of 18.52, 17.46, 17.70 for F336W, F475W, and F814W, respectively. In comparison, the average integrated magnitudes of the added backgrounds for the three passbands are 16.51, 15.20, 15.28, respectively. In addition, most mock images correspond to N -body simulations, which are equivalent to low-density Galactic GCs (Bianchini et al. 2026). Their lower surface brightness, and consequently lower S/N, likely makes recovering their intrinsic properties more challenging.

For brighter clusters (~ 12 – 17 mag), the errors are smaller, especially for age, ellipticity, and position angle, while the distance errors are roughly independent of brightness. In contrast, the total mass is typically overestimated for brighter GCs while the decontamination in F475W and F814W shows a nearly constant brightness underestimation (e.g. bias) which on average is 0.43 mag and 0.64 mag in F475W and F814W, respectively. These discrepancies become more severe for both really faint and really bright clusters, but is nonetheless present across the entire range. There is a similar shift of 0.31 mag also for the F336W passband. The overall uncertainty thereby increases with redder passbands: F336W is best constrained and F814W the worst. Because the brightness in each passband is underestimated by different amounts, it also means that the colours of the decontaminated GC are inconsistent, with π -DOC predicting bluer colours on average.

These intrinsic errors are also comparable to limitations in the field, and we regard the 2σ limits as a conservative confidence range for π -DOC's predictions (not to be confused with the 1σ limits indicated in the figure). For the F336W, F475W, and F814W passbands, the decontamination errors are confined to $^{+0.97}_{-0.07}$, $^{+1.16}_{+0.06}$, and $^{+1.48}_{+0.12}$ mag, respectively. The mass ($^{+0.23}_{-0.50}$ dex) and age ($^{+2.86}_{-3.46}$ Gyr) are well constrained compared to typical uncertainties for M31 GCs (Fan et al. 2010; Chen et al. 2016; Usher et al. 2024). This mass confidence limit is only around a factor 1.5 as wide as typical uncertainties for fully resolved Galactic GCs (see e.g. Bellini et al. 2017; Zocchi et al. 2017). The lower limit of the mass errors is further exacerbated by the MOCCA models, which are consistently underestimated (see the separate island in the bottom part of the corresponding panel in column a). Moreover, a handful of outliers lie beyond the 2σ boundary, particularly for the brightest clusters where masses are underestimated by factors exceeding four; these rare cases mainly correspond to early snapshots ($\lesssim 200$ Myr) from the underrepresented high-mass regime ($\gtrsim 10^6 M_\odot$) of the MOCCA simulations.

Distance ($^{+77}_{-80}$ kpc), ellipticity ($^{+0.06}_{-0.09}$), and particularly P.A. ($^{+143^\circ}_{-130^\circ}$) show larger uncertainties. The confidence range for the ellipticity and P.A. are primarily exacerbated by faint, low-S/N clusters which is evident from the elongated contours in column (a) of Figure 6. Excluding clusters fainter than 18 mag (F336W) and with an ellipticity below 0.05 (where the P.A. is rather ill-defined) yields a more robust confidence range for the P.A. ($^{+44^\circ}_{-19^\circ}$) but still indicates that there is an overall difficulty in inferring a GC's P.A. (even though the 1σ are well constrained).

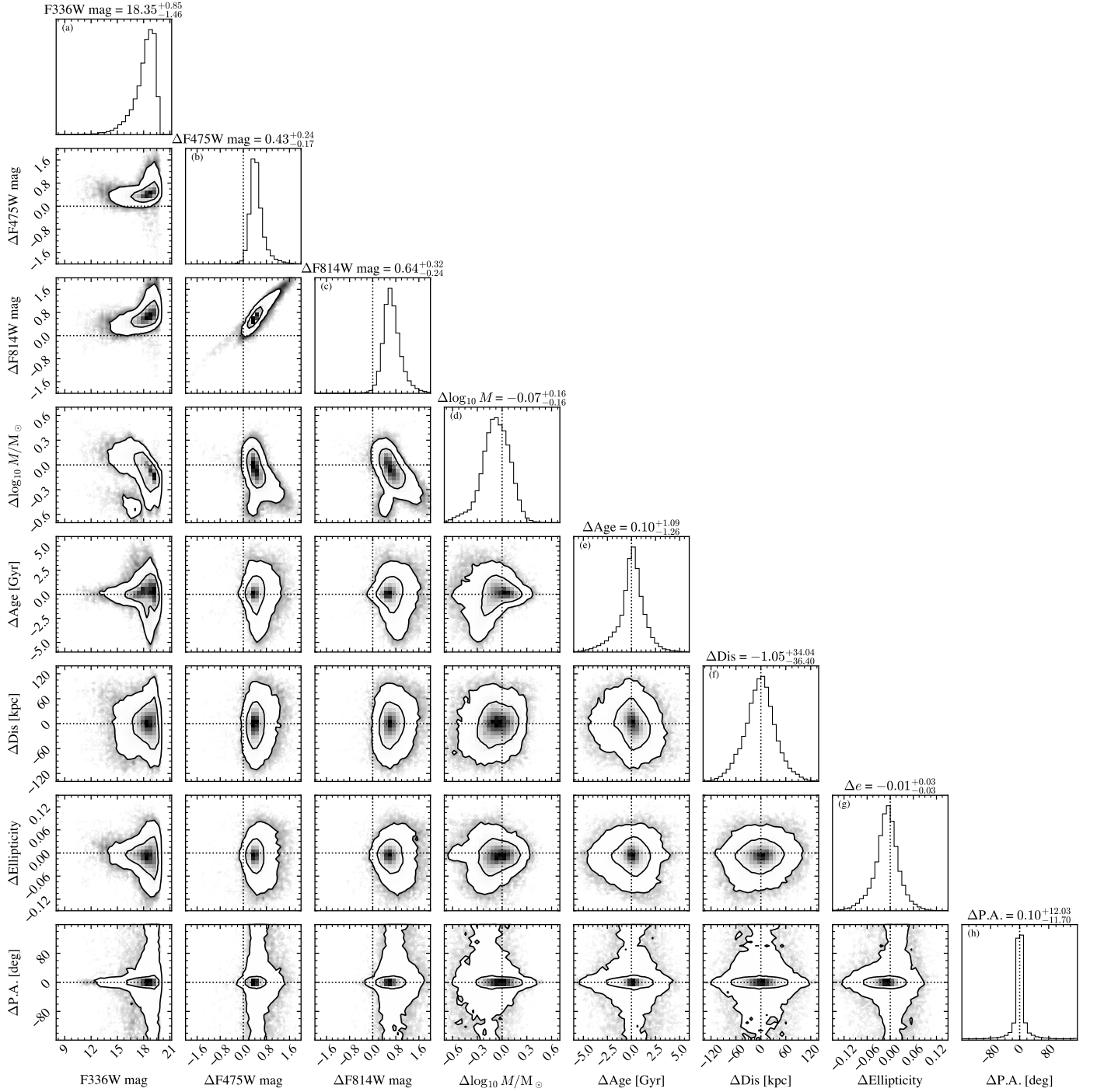


Fig. 6. Corner plot showing the distributions and correlated distribution of the errors of the main predicted outputs of π -DOC. The contours are set to σ , 2σ , and the dotted lines indicate the 0-axes (e.g. no discrepancy), where applicable.

We also include a similar exploration of biases and uncertainties with respect to the relative GC-to-background brightness that more explicitly targets the S/N in Appendix C. Overall, we find a similar dependence on S/N as we do for the GC’s brightness. Moreover, π -DOC demonstrates an impressive resilience to low S/N (see in particular Figure C.2). Although this is beyond the purposes of this study, these results suggest π -DOC could be adapted to simultaneously detect GCs (Barbisan et al. 2022; Dold & Fahrion 2022; Singlow et al. 2022) in addition to inferring their properties.

5. Reliability of predictions and representational similarity

In this section, we consider two complementary approaches for assessing the reliability of π -DOC’s predictions. The first estimates the network’s own confidence in a given prediction (Section 5.1); the second assesses how representative an input is of the training data itself, via the Mahalanobis distance (Section 5.2).

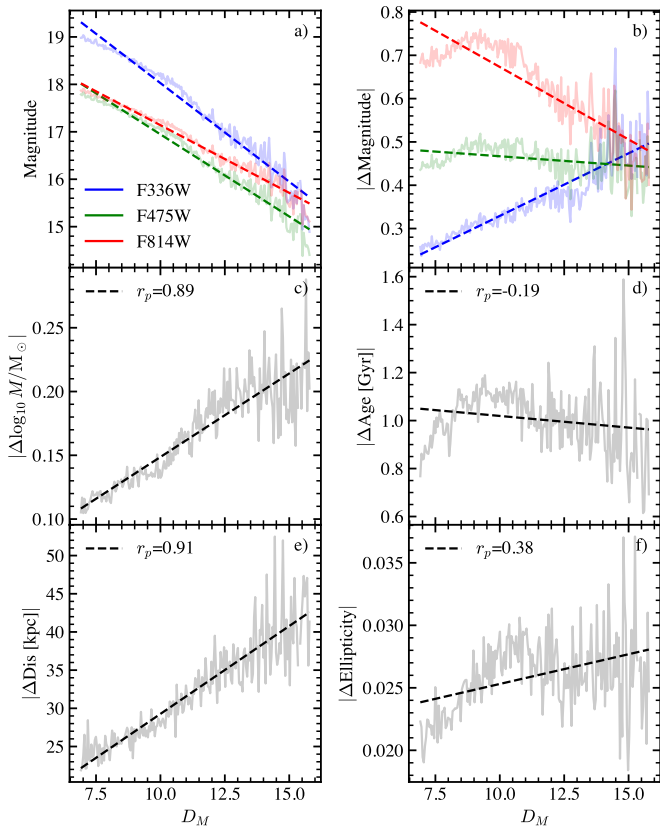


Fig. 7. Dependence on D_M for the total integrated magnitudes (panel a) with corresponding errors in panel (b), errors for total mass (panel c), errors for age (panel d), error for distance (panel e), and error for ellipticity (panel f). We find a strong linear correlation between D_M and the quantities in left-hand side column, while those in the right-hand side column show non-linear and weaker correlations.

5.1. Epistemic uncertainty

Estimating a network’s uncertainty for individual predictions in a reliable manner remains challenging. To quantify a predictive uncertainty in our model, we adopt Monte Carlo (MC) dropout, a computationally efficient approximation to Bayesian inference (see Gal & Ghahramani 2015a,b; Gal et al. 2017). During training, dropout layers randomly deactivate subsets of neurons to prevent overfitting. In MC dropout, this stochastic behavior is retained during inference by performing multiple forward passes with dropout enabled. Each forward pass samples a different realisation of the network’s weights, effectively drawing from an approximate posterior distribution over models. For a given input, we obtain an ensemble of predictions that provide a measure of the model’s so-called epistemic uncertainty.

For the majority of estimates, the epistemic uncertainties remain small and do not reflect the differences of the predictions to the ground truth. These values should not be regarded as physical uncertainties, but rather as the intrinsic confidence level of π -DOC, and we include them in our final catalogue and in the forthcoming sections. More details on our implementation can be found in Appendix D.

5.2. Mahalanobis distance

π -DOC is trained exclusively on synthetic data generated under controlled physical assumptions, therefore, when applied to real astronomical observations, there is no guarantee that all inputs

lie within the same statistical distribution as the training set. Predictions made on inputs that differ significantly from the training distribution may therefore be unreliable, even if the model reports low internal uncertainty. A common approach is to define an out-of-distribution (OOD) classification by thresholding the Mahalanobis distance (D_M) between a target’s latent-space representation and the training distribution, typically calibrated to a fixed false-positive rate (FPR) on the training sample itself (Lee et al. 2018; Gonzalez et al. 2022). This binary classification is attractive for its simplicity, but it is inherently arbitrary: the training distribution has no natural density edge.

We therefore treat D_M as a continuous indicator of representational similarity to the training data, rather than as a binary classification based on an arbitrary threshold, and ask whether D_M traces the estimate quality: if the latent space organises clusters in a physically meaningful way, clusters further from the training manifold should show systematically larger discrepancies from the ground truth.

We extract intermediate feature representations from the latent space (see Fig. 2), keeping the features agnostic to specific output heads. These feature vectors are projected onto a lower-dimensional subspace using incremental principal component analysis (see e.g. Abdi & Williams 2010) to improve numerical stability and reduce redundancy. Assuming that the feature distribution of the training data can be approximated by a multivariate Gaussian in this low-dimensional space, we characterise it by its mean vector and covariance matrix. Given a new input, we compute D_M between its feature representation and the training distribution.

We find that D_M correlates strongly and linearly with the intrinsic brightness of the GC (Figure 7, with Pearson coefficients of $r_p \sim -0.9$ to -0.99 across passbands), indicating that the dominant axis of variation captured by this representation is closely tied to GC brightness. D_M also correlates with the brightness of the background and with the relative GC-to-background contrast, though these relationships are non-linear, exhibiting local extrema rather than a monotonic trend; we do not interpret these secondary trends further, as they likely reflect the combined influence of several correlated observational properties (GC brightness, background brightness, S/N, and plausibly other physical properties).

This provides a natural, if partial, explanation for several of the correlations seen in the remaining panels of Figure 7, which shows D_M against the errors (discrepancy between predicted and ground truth values, see Section 4.2) for the total integrated magnitudes (panel b), total mass (panel c), age (panel d), distance (panel e), and ellipticity (panel f). Mass and distance are, like brightness, strongly and linearly correlated with D_M , and are therefore more accurately recovered at low D_M . Age and ellipticity, by contrast, show little correlation with D_M ($|r_p| \lesssim 0.2$ and $|r_p| \lesssim 0.4$, respectively); for age, this is plausibly related to a multi-dimensional dependence on colour, morphology, and metallicity rather than on brightness alone, while ellipticity is not expected to relate strongly to either and also shows very small overall errors.

The decontamination errors (Δ Magnitude, panel b) show a more intriguing pattern: while D_M traces GC brightness comparably across all three passbands, the associated errors do not respond uniformly (nor strictly linearly either). The F336W error increases with D_M , while the F814W error decreases with a comparable slope of opposite sign; F475W lies in between and is only weakly dependent on D_M . We do not consider this a coincidence, but we are cautious not to over-interpret it: one plausible explanation is a colour-dependent bias in the decontamination (pre-

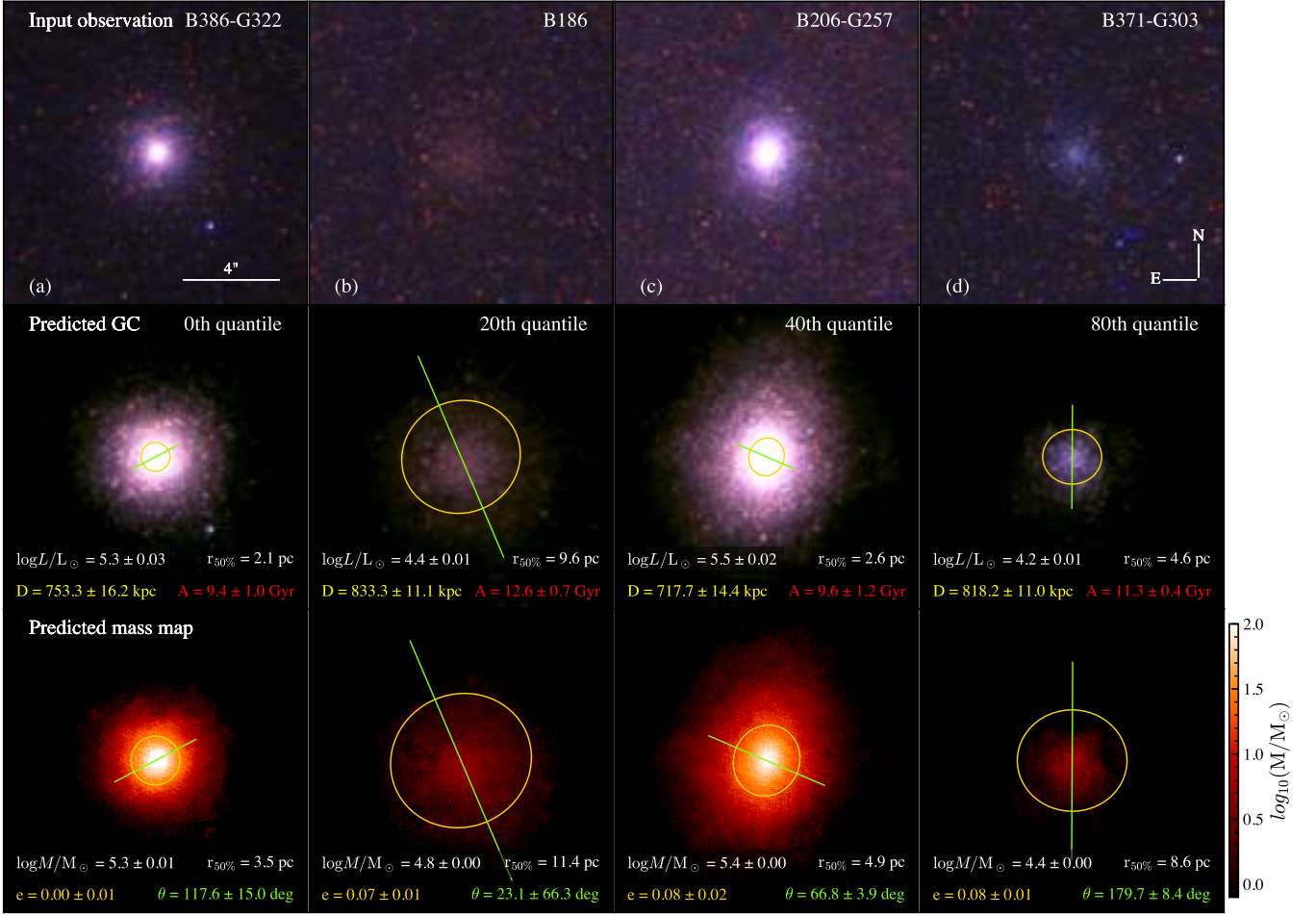


Fig. 8. Mosaic of four selected M31 GCs in the PHAT survey from left to right: Bo1 386, Bo1 186, Bo1 206, and Bo1 371. First row: input observations as RGB images scaled identically to the inputs shown in Figure 4. Second row: corresponding decontaminated GC maps, also shown as RGB images and scaled identically to the ground truth and decontaminated maps in Figure 4. The total estimated luminosities and half-light radii corresponds to the F814W passband and are obtained from the corresponding decontaminated image. Third row: estimated mass maps with derived total mass and half-mass radii. The golden ellipses indicate the estimated ellipticity, and its major-axis extends to the half-light or half-mass radius, while the green line cutting through the ellipse indicates the position angle of the minor axis. The estimated values for the ellipticity and the position angle are indicated in the bottom row, while the estimates for age and distance are indicated in the middle row. The epistemic uncertainty for each scalar quantity is also indicated.

viously suggested in Section 4.2 where magnitudes in different passbands appear to be shifted towards each other, producing smaller colour differences). Indeed, we find that D_M correlates strongly and approximately linearly with GC colour and with the GC-background colour contrast (in all cases $|r_p| \gtrsim 0.9$), with bluer clusters and lower colour contrast found at higher D_M .

Given the multi-dimensional nature of the latent representation from which D_M is derived, we caution against interpreting these individual correlations as evidence that the underlying quantities are directly related to one another: two properties that each correlate with D_M need not be correlated with each other, and disentangling their respective contributions to D_M is beyond the scope of this work.

To conclude, D_M is best understood as a proxy for similarity to the training manifold. We therefore regard D_M as a useful indicator of the reliability of our predictions, and more strongly for outputs that depend cleanly on a single property D_M traces well (brightness, colour).

6. Application to M31 GCs

In this section we infer properties for 349 M31 GCs using π -DOC. Following the outline of Section 4, we begin by demonstrating π -DOC's performance on four example GCs in Section 6.1 and compare their inferred values to independent studies. In Section 6.2 we apply π -DOC on all GCs within the PHAT and PHAST footprints, using π -DOC₃ and π -DOC₂, respectively. We then compare our findings to four independent studies in Section 6.3. The inference speed of π -DOC is ~ 10 GCs s⁻¹, so the full M31 dataset of 349 GCs required only about half a minute¹⁹. We provide an online repository²⁰ containing decontaminated images, mass maps, and all remaining predicted quantities, including epistemic uncertainties and D_M (for a reliability and similarity-score to the training data) for all GCs con-

¹⁹ With MC dropout enabled, the corresponding speed is ~ 0.7 s per GC for π -DOC₃ and ~ 0.5 s per GC for π -DOC₂ TEST ON MY MAC

²⁰ Zenodo link will go here

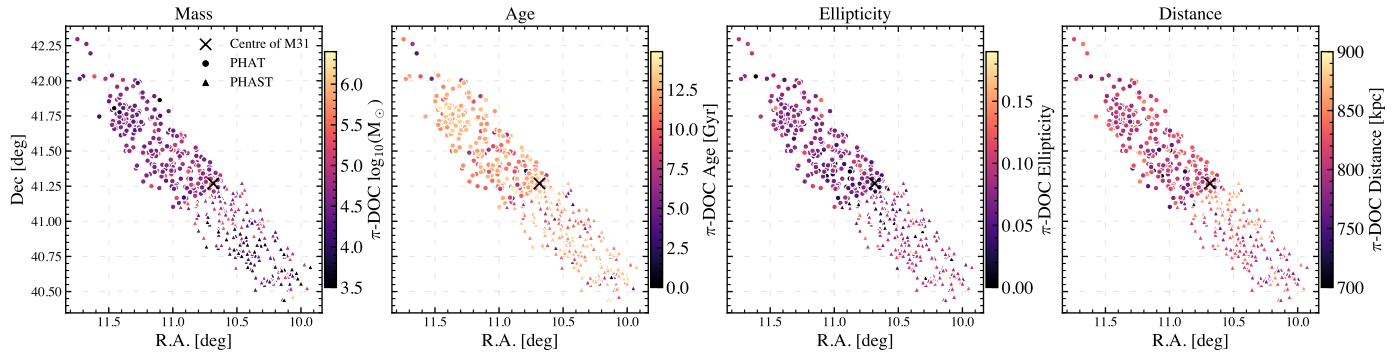


Fig. 9. GCs within the footprints of the PHAT (circles) and PHAST (triangles) surveys, colour-coded by π -DOC estimates of total mass (a), age (b), ellipticity (c), and distance (d). OOD clusters are outlined in white. There is a tentative concentration of higher masses toward the centre, but OOD flags necessitate cautious interpretation. The oldest GCs according to π -DOC lie in the north-east region of M31, which also corresponds to larger inferred distances; this trend lacks a clear physical explanation and likely reflects biases in π -DOC and observational systematics such as extinction and age–metallicity gradients. Most GCs appear nearly circular, with the highest ellipticities typically found at larger galactocentric distances, likely reflecting S/N.

sidered in the PHAT and PHAST surveys. The results are also reported in Table E.1.

6.1. Mosaic example of four GCs

The four examples shown in Figure 8 have been chosen based on their D_M to the training manifold: (a) B386–G322 has the smallest D_M , (b) B186 lies at the 20th quantile, (c) B206 lies at the 40th quantile, and (d) B371–G303 lies at the 80th quantile. Each GC is located in the PHAT survey, and hence we employ π -DOC₃. Despite the four clusters representing examples of decreasing similarity with respect to the training data, they do not exhibit any visually striking degradation in the reconstructed mass maps, decontamination quality, or in the values returned by the regression head. This suggests that increasing D_M does not necessarily manifest as obvious artefacts or implausible predictions in the individual outputs.

Comparing the predicted masses and ages with the three independent literature catalogues (Fan et al. 2010; Chen et al. 2016; Usher et al. 2024) highlights both an overall consistency and an expected scatter.

- For B386–G322, which is available in all three catalogues, our mass estimate ($\log M/M_\odot = 5.31$) lies between the values reported by Fan et al. (2010, $\log M/M_\odot = 5.28$), Chen et al. (2016, $\log M/M_\odot = 6.47$), and Usher et al. (2024, $\log M/M_\odot = 5.89$). The corresponding age estimates show clear disagreement between the studies; π -DOC’s value of 9.43 Gyr agrees only with Usher et al. (2024) (8.26 Gyr), versus 1.28 Gyr from Fan et al. (2010) and 17.39 Gyr from Chen et al. (2016).
- For B186, which is only present in Fan et al. (2010) and Chen et al. (2016), our mass ($\log M/M_\odot = 4.78$) is lower than both literature values ($\log M/M_\odot = 5.40$ and 5.39 , respectively), while the age (12.64 Gyr) is similar to Fan et al. (2010, 12.75 Gyr) and older than Chen et al. (2016, 5.88 Gyr).
- For B206, our mass ($\log M/M_\odot = 5.43$) is again lower than the literature estimates ($\log M/M_\odot = 5.62$, 6.21 , and 6.10 in Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024), respectively), whereas for the age we again only find an agreement with Usher et al. (2024) (9.58 Gyr vs 1.61 Gyr, 5.01 Gyr, and 9.74 Gyr).
- The fourth object, B371–G303, has no counterpart in these three catalogues, so no direct comparison is possible.

Expressed as linear mass ratios, the literature-to- π -DOC factors for the three clusters span roughly a factor of 1.5 to 15, with the largest discrepancy for B386–G322 against Chen et al. (2016) (factor ~ 14.6) and more moderate differences for the other cases (factors ~ 1.0 – 6.0). Alongside the analysis of the performance on the synthetic data (Section 4), it seems likely that π -DOC is underestimating these masses, and particularly for B386–G322 and B206–G257.

The inferred distances span more than 100 kpc, from ~ 718 kpc (B206–G257) to ~ 833 kpc (B186), and appear to correlate with cluster brightness: the brighter B206–G257 and B386–G322 lie at smaller distances, while the fainter B186 and B371–G303 are placed further away. However, this trend is not conclusive. If the more distant clusters were simply on the far side of M31, one might expect higher extinction, yet Fan et al. (2010) find similar $E(B - V)$ for B186 and B386 (0.24 and 0.21 mag).

The clusters show a small morphological diversity. B386–G322 is spherical, while the others have ellipticities of $e \sim 0.07$ – 0.08 . For B206–G257, we find close agreement with Staneva et al. (1996, 0.085 vs. 0.086), whereas for B386–G322 we do not (~ 0 vs. 0.10). For B206–G257, the position angle differs by roughly 40° (67° vs. 24°), which is significant; however, this discrepancy is of limited concern given the cluster’s low ellipticity, for which the position angle is only weakly constrained.

6.2. Inference of GC properties within the PHAT and PHAST footprints

Figure 9 shows the GCs within the PHAT and PHAST footprints, colour-coded by the mass (a), age (b), ellipticity (c), and distance (d), as estimated by respective π -DOC model. Overall, we find that the mass distribution is relatively uniform, though more massive GCs tend to lie closer to the M31 centre (on average, there is a decrease of $\sim 1200 M_\odot$ per arcminute from the centre). This could be interpreted as a consequence of orbital decay driven by dynamical friction, but it may also reflect the intrinsic uncertainties of π -DOC and the aforementioned biases we have discussed in previous sections.

Most of the GCs are old (≥ 10 Gyr) and appear relatively circular, with only a few exceeding an ellipticity of 0.1; these are predominantly located well outside the central regions of M31. This might possibly reflect a bias in underestimating ellipticity

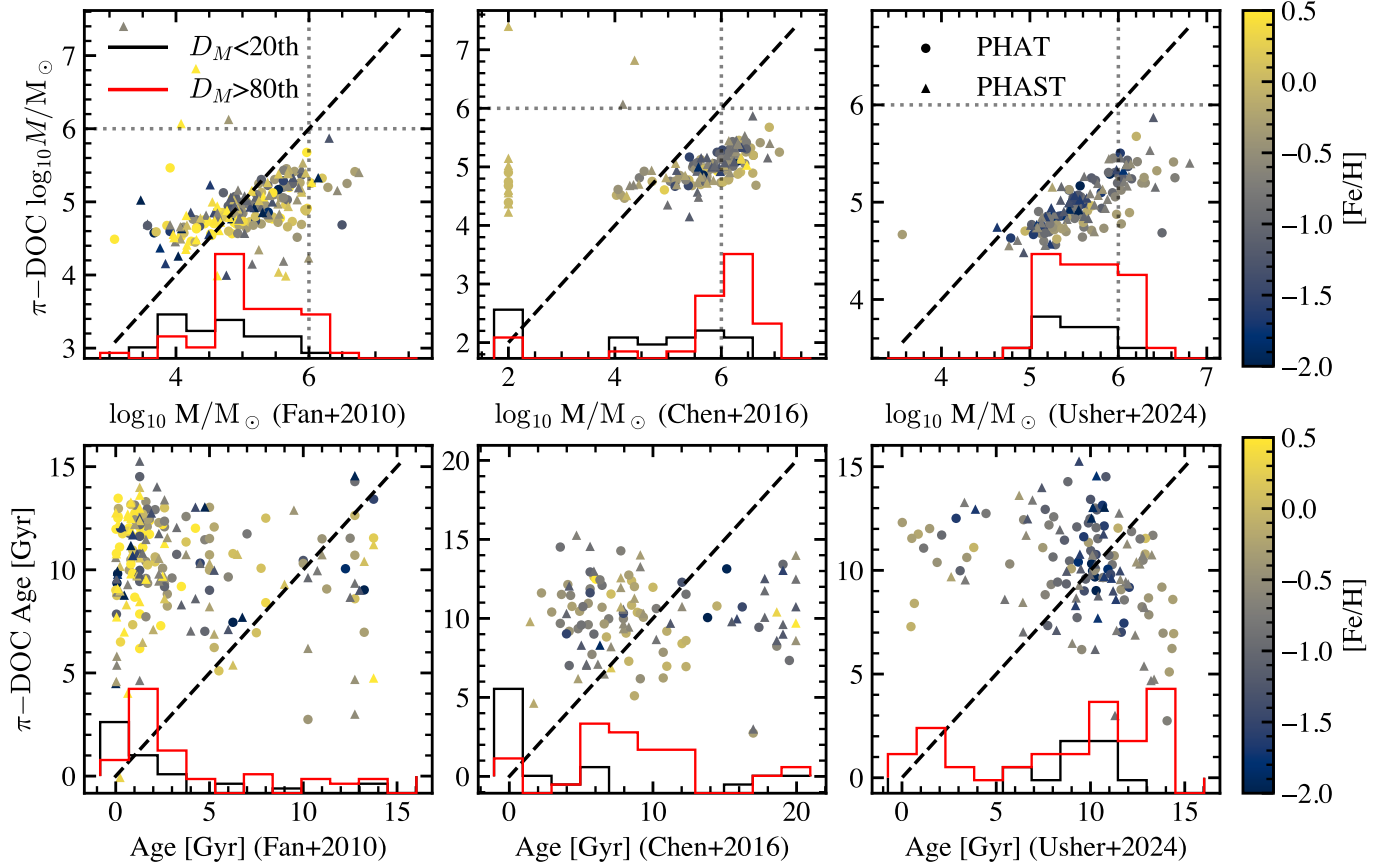


Fig. 10. Comparison of the π -DOC mass and age estimates with those of [Fan et al. \(2010\)](#), [Chen et al. \(2016\)](#), and [Usher et al. \(2024\)](#), for 206, 143, and 141, GCs, respectively. The dashed black line indicates a 1:1 relation, and the dotted lines mark $10^6 M_\odot$, above which reliable coverage is not expected (see Fig. 1). Points are colour-coded by the metallicity estimates from respective study. We find overall agreement in total mass to within a factor of about four, while acceptable age agreement is obtained only for low-metallicity GCs with [Usher et al. \(2024\)](#). The black histogram corresponds to GCs within the 20th quantile and the red histogram corresponds to GCs outside the 80th quantile of D_M .

measurements for GCs at lower S/N (see Section 4.2), i.e. closer to the central regions.

We find an average distance 813 ± 31 kpc to the GCs in M31, with a slight difference between PHAT (805 ± 27 kpc)²¹ and PHAST (825 ± 33 kpc). A direct comparison of the networks' estimates for PHAT GCs is also included in Appendix B, which confirms overall agreement while showing that π -DOC₂ typically infers greater distances than π -DOC₃ for the same input (see Figure B.1).

6.3. Mass and age literature comparisons

We proceed to compare the full sample of GCs to the three independent studies in Figure 10. For each respective study, we have an overlapping sample of 206 ([Fan et al. 2010](#)), 143 ([Chen et al. 2016](#)), and 141 ([Usher et al. 2024](#)) GCs, respectively.

We find that the comparisons in Section 6.1 were fairly representative of the overall agreement with these independent studies. For the mass, we find a clear rank-order agreement: clusters identified as more massive in the literature are also assigned higher masses by π -DOC, and vice versa. However, there is a systematic offset in the absolute scale, with π -DOC's masses increasingly lower than literature values at the high-mass end

($\gtrsim 10^6 M_\odot$). This is particularly clear for individual cluster comparisons with [Chen et al. \(2016\)](#) and [Usher et al. \(2024\)](#). It is, furthermore, reflected in the average GC mass ($\log_{10}(M/M_\odot) = 4.72 \pm 0.5$ for our full sample and $\log_{10}(M/M_\odot) = 4.96 \pm 0.36$ for overlapping sample) which is a factor of 1.3-4 $\times$ lower than the corresponding mean mass derived from the overlapping samples with [Fan et al. \(2010\)](#), [Chen et al. \(2016\)](#), and [Usher et al. \(2024\)](#). This is also in line with the biases seen for the synthetic dataset tests in Sections 4 and 4.2 (see also Figure 5), where the most massive clusters are underestimated up to a factor of around four.

For the ages, we also find a fairly good agreement with the overall distribution of [Usher et al. \(2024\)](#), whereas we find a very poor agreement with [Fan et al. \(2010\)](#) and [Chen et al. \(2016\)](#). We note that there is not a clear agreement among the three independent studies either.

6.3.1. D_M as a proxy for prediction reliability

In order to address the reliability of our estimates for real GCs, we conduct a similar exploration of the correlation with D_M to that presented in Section 5. We consider the differences in mass and age estimates between π -DOC and [Fan et al. \(2010\)](#), [Chen et al. \(2016\)](#), and [Usher et al. \(2024\)](#).

²¹ When employing π -DOC₂ on the PHAT survey, we find (820 ± 32 kpc).

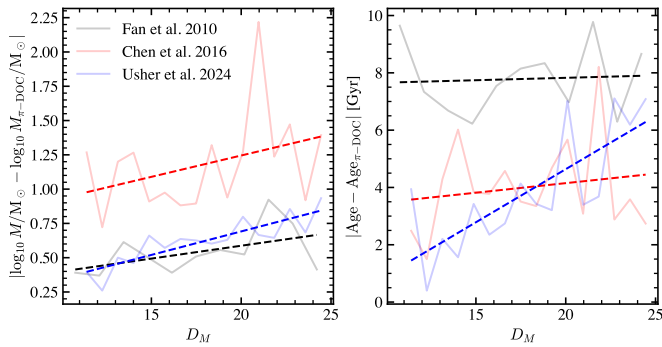


Fig. 11. Absolute differences in mass estimates (left panel) and age estimates (right panel) between π -DOC and Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024), as a function of the Mahalanobis distance D_M .

Each panel in Figure 10 further indicates the distribution of GCs below the 20th percentile and above the 80th percentile of D_M , representing the most similar and least similar GCs to the training data, respectively. The least similar GCs are shifted towards higher masses, in line with the poor training-set coverage of very massive GCs discussed above. For Usher et al. (2024) there is, in addition, a much more narrower distribution for the age (roughly centred at the region where we find the most agreement). The most pronounced shift, however, occurs for metallicity ([Fe/H]): for Fan et al. (2010), the below-20th-percentile and above-80th-percentile groups have means and standard deviations of -0.64 ± 0.89 and -0.20 ± 0.73 dex, respectively; for Chen et al. (2016), -0.58 ± 0.57 and -0.31 ± 0.32 dex; and for Usher et al. (2024), -1.20 ± 0.48 and -0.72 ± 0.43 dex. In other words, the least similar GCs are systematically more metal-rich. This likely also explains our comparatively better agreement for age with Usher et al. (2024), whose inferred metallicities are closer to those of our training data ([Fe/H] ~ -1.3 ; Askar et al. 2025; Zhao et al. 2026; Bianchini et al. 2026; Marklund et al. 2026, see Section 2.2).

Figure 11 further supports the discussion above. We find that agreement with literature values improves continuously with decreasing D_M across the full sample, for all three comparisons. The corresponding Pearson correlation coefficients between D_M and the discrepancy with π -DOC are $r_p = 0.49$ (mass) and 0.06 (age) for Fan et al. (2010); 0.36 (mass) and 0.17 (age) for Chen et al. (2016); and 0.85 (mass) and 0.76 (age) for Usher et al. (2024). This reinforces D_M as a proxy for estimate reliability on real data, though we note one notable difference from the synthetic case: age discrepancy correlates strongly with D_M for Usher et al. (2024) (and, more weakly, if at all, for the other two studies), whereas no such relationship was found for age on the synthetic test set.

Since the real and synthetic comparisons differ both in the underlying data distribution and in what is being measured (discrepancy with an independent literature estimate, rather than error against known ground truth), we do not consider this a contradiction; it may instead reflect physical correlations that are not represented in our synthetic training grid. We leave a more detailed investigation of this discrepancy to future work.

6.3.2. Methodological discrepancies

Part of the discrepancies with the literature studies may also arise from differences in methodology and observational data. Usher et al. (2024) combined ground-based photometry from the

Sloan Digital Sky Survey (SDSS; York et al. 2000) and the Pan-Andromeda Archaeological Survey (PAndAS; Huxor et al. 2014), and with spectroscopy from Caldwell et al. (2009); Sakari & Wallerstein (2016, 2022); Sakari et al. (2021) where they target the CaT region.

Using Markov Chain Monte Carlo (MCMC), they simultaneously fitted ages, metallicities, present-day stellar masses, and reddening with FSPS models (Conroy et al. 2009; Conroy & Gunn 2010) based on MIST isochrones (Choi et al. 2016). Explicit reddening modeling systematically increases the inferred total brightness and thus the associated mass estimates compared to our values, since we only account for Milky Way reddening. At the same time, their photometric masses trace the current surviving stellar populations and do not include dynamical mass loss (e.g. stellar-mass dependent evaporation and stripping), which π -DOC does; photometric and dynamical mass estimates are known to show discrepancies (Sollima et al. 2017).

Chen et al. (2016) also combined ground-based photometry from Peacock et al. (2010a) with spectroscopy using the Large Sky Area Multi-Object Fibre Spectroscopic Telescope (LAMOST Liu et al. 2015; Yuan et al. 2015). Ages and metallicities were derived via pixel-by-pixel fitting of integrated spectra to single stellar population (SSP) models, with masses inferred from multi-band photometric spectral energy distributions (SEDs) scaled to those parameters. Similarly, Fan et al. (2010) also derived age and mass estimates by comparing their observed SEDs with SSP models. Their estimates thus yield photometric masses and are subject to the same biases as Usher et al. (2024).

We have already mentioned the metallicity distribution as a likely cause for the poor overall agreement for age. However, studies that agree on metallicity, such as Chen et al. (2016) and Usher et al. (2024), exhibit comparable scatter to that seen with π -DOC, demonstrating that all literature estimates show discrepancies (see Usher et al. 2024). The presence of ages near 0 Gyr or 20 Gyr in Fan et al. (2010) and Chen et al. (2016) further underscores the intrinsic challenges of GC age determination. These comparisons therefore demonstrate that age estimation remains notoriously difficult across methods, rather than indicating specific shortcomings of π -DOC or other studies specifically. Importantly, the correlation between metallicity and D_M we identified via the comparisons of the 20th and 80th quantiles of the D_M distribution, suggests that metallicity information could already be encoded in the network’s features. This opens the possibility of extending future versions of π -DOC to jointly predict age and metallicity as explicit outputs, thereby providing a new approach to breaking the age-metallicity degeneracy.

6.4. Literature comparisons of GCs’ morphologies

We compare our ellipticity and position angle estimates with Staneva et al. (1996) in Figure 12, where we have a total overlapping sample of 103 GCs. There is a broad agreement in the overall ellipticity distribution, but π -DOC generally infers lower values. Discrepancies are to be expected given the differences in methodology: Staneva et al. (1996) derived ellipticities from iso-density contours of ground based photometry, whereas π -DOC was trained on the intrinsic stellar distributions within the field of view, quantified using the second-moment tensor method (see e.g. Marklund et al. 2026, for details). These two approaches are known to produce systematically different results even for identical stellar distributions (Fréour et al. 2026; Marklund et al. 2026). Furthermore, Fréour et al. (2026) demonstrated that ellipticity measurements for more spherical distributions ($e \lesssim 0.1$) are known to produce large systematic overestimates and that

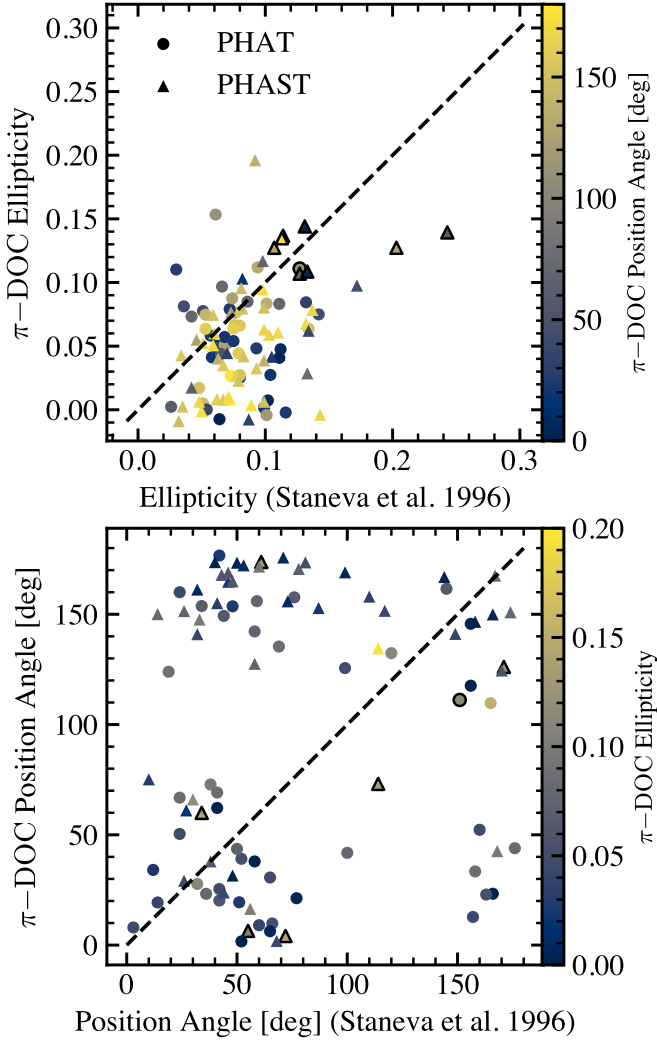


Fig. 12. Comparison between π -DOC’s ellipticity and position angles estimates for 103 GCs to those in Staneva et al. (1996). π -DOC’s ellipticity estimates are systematically lower than those of (Staneva et al. 1996), and the position angles show a poor overall agreement. This can partially be explained by the degeneracy at 0° , 180° and overall small values for ellipticity. The dashed black line indicates a 1:1 relation.

these biases need to be corrected, which is not done for (Staneva et al. 1996).

A second important factor is spatial resolution and coverage. Ground-based observations inevitably have lower resolution and typically a wider FoV than HST imaging. Outer cluster regions, which dominate ground-based measurements, are generally more elliptical than inner regions (Fréour et al. 2026; Marklund et al. 2026). For spatially resolved clusters in the Milky Way, Fréour et al. (2026) found that typically there is not a significant difference between the measured flattening within $3r_{50\%}$ and $19r_{50\%}$. On the contrary, the intrinsic flattening within and outside $r_{50\%}$ may show significant differences (Marklund et al. 2026). Although our mock images extend beyond the half-mass radius of the clusters, they remain confined to radii of approximately 20 – 30pc (set by the chosen FoV of our images and subject to the distance). This difference in spatial resolution and radial coverage may naturally contribute to the observed offsets.

Although direct one-to-one agreement in ellipticity across different studies should not be expected (Fréour et al. 2026), one

might at least anticipate reasonable agreement in position angle for clusters identified as elliptical ($e \geq 0.1$) by both Staneva et al. (1996) and π -DOC. These points are highlighted with black edges in Fig. 12, and show that differences exceeding $20\text{--}30^\circ$ persist even in this regime. We attribute this to the position angle being inherently poorly constrained and sensitive to measurement choices — such as isophote level, background subtraction, and fitting method — any of which can shift the inferred orientation even when the underlying structure is similar (Fréour et al. 2026; Marklund et al. 2026).

7. Improving performance on massive GCs

We have suggested throughout this work that π -DOC is less reliable for the most massive GCs, primarily due to their limited representation in the training data — particularly for the MOCCA models. If this is correct, a network exposed to more such clusters should show improved performance in the high-mass regime ($\geq 10^6 M_\odot$). To test this, we continue training π -DOC₃ for 10 additional epochs (~ 300 GPU hours) with two changes: (i) all loss weights (Section 3.4) are set to zero except decontamination ($w_{\text{decontamination}} = 1$) and mass ($w_{\text{mass}} = 10$), so the network focuses solely on these two outputs (and should not be employed for other quantities), contributing roughly equally to the total loss (Table 1); and (ii) each batch is constrained to include at least 25% MOCCA models, increasing their relative weight in training. We use a low learning rate (10^{-7}) to minimise over-fitting (given the limited number of MOCCA models, each batch become more alike). We denote the resulting network π -DOC₃^{MOCCA}, and stress that this is a proof-of-concept test rather than an optimised configuration: adjusting these hyperparameters further is likely to improve both performance and convergence speed.

Figure 13 shows that this fine-tuning shifts the predicted MOCCA masses upward (see panel a) and reduces the underestimation bias (see panel b) across all models, including the N -body ones. This latter improvement likely reflects the network training exclusively on decontamination and mass, rather than the increased MOCCA fraction specifically. For the synthetic test set, the recovered mean and 1σ of the mass distribution is $5.28^{+0.06}_{-0.08}$ dex for π -DOC₃ and $5.45^{+0.07}_{-0.08}$ dex for π -DOC₃^{MOCCA} (ground truth: $5.77^{+0.03}_{-0.05}$ dex). As a result, the systematic underestimation in the MOCCA models is reduced from a factor of 3 to a factor of 2. A similar shift is reflected in the PHAT sample (see Section 6 and in particular Section 6.3): $4.70^{+0.06}_{-0.13}$ dex for π -DOC₃ versus $4.82^{+0.05}_{-0.16}$ dex for π -DOC₃^{MOCCA}.

We also argued that a possible, and not mutually exclusive, cause for the discrepancies were related to the MOCCA models having a more centrally concentrated population of stellar remnants that resulted in a fundamentally different M/L radial dependence. Figure 14 shows that this is the case: (i) both the N -body and MOCCA models are dominated by remnants in their central regions, but the ratio between the remnant and visible mass components is markedly more extreme for the MOCCA model; (ii) π -DOC’s predictions closely track the visible mass rather than the total mass, so the underestimated mass is concentrated precisely in the remnant-dominated regions; (iii) the improved mass estimate on the MOCCA model using π -DOC₃^{MOCCA}, although marginal, is applied roughly uniformly across radius rather than being concentrated in the remnant-dominated core — suggesting the fine-tuning corrects the network’s overall visible-to-total mass scaling rather than specifi-

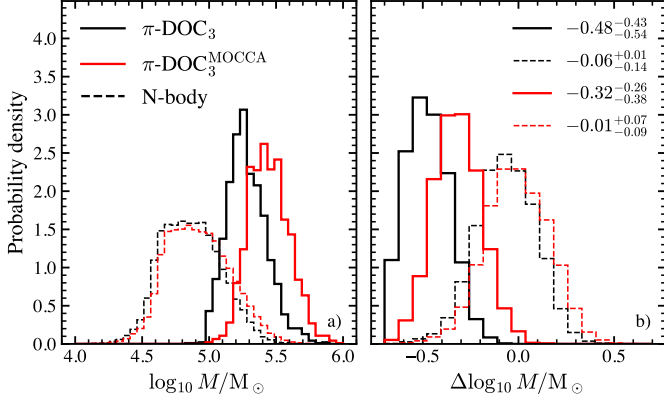


Fig. 13. Recovered probability density distribution for the mass (panel a) and corresponding errors (panel b) for MOCCA (solid lines) and N-body (dashed lines) mock images using π -DOC₃ (solid black line) and π -DOC₃^{MOCCA} (solid red line). The labels in panel (b) indicate the mean and $\pm 1\sigma$ of corresponding error distribution.

cally learning to identify the remnant-dominated regions themselves.

This is expected, since π -DOC infers mass from the observed light distribution and has no direct observational handle on non-luminous remnant mass; it can therefore only apply an implicit, statistically learned correction calibrated to the training set’s typical remnant concentration. This also explains why the MOCCA-focused fine-tuning above improves, but does not fully resolve, the high-mass underestimation: the correction it learns remains an average over the training distribution rather than a direct measurement of the (invisible) remnant population in any individual cluster. A promising extension would be to explicitly condition the pixel-wise mass correction on global cluster properties (e.g. via attention or feature-wise modulation), allowing the network to apply a context-dependent M/L correction.

These results confirm that increasing the relative fraction of MOCCA models in training improves performance in the high-mass regime, and suggest more generally that π -DOC can be fine-tuned toward specific tasks — for example, improving the distance estimates, which we know to be biased toward the mean of the training distribution (see Figure 5). For this paper, we prioritised a general-purpose model with balanced performance across all outputs, but future versions could exploit this targeted fine-tuning approach. This targeted approach also suggests a more exploratory possibility. Several of our MOCCA models host intermediate-mass black holes; a similarly fine-tuned network could, in principle, be trained to specifically flag or search for their signatures (Pechetti et al. 2022). We do not pursue this here, and stress it as a speculative extension rather than a demonstrated capability of the current framework — but see it as an intriguing avenue for future work.

8. Conclusions

In this work, we introduced two complementary frameworks: CREMANT, a forward-modelling framework that transforms simulation snapshots into realistic mock observations (see Section 2), and π -DOC, a neural network that infers intrinsic dynamical and morphological GC properties from multi-band photometric images and pixel CMDs (see Section 3). Both were developed using HST observations of M31 from the PHAT and PHAST surveys (Dalcanton et al. 2012; Chen et al. 2025). π -DOC performs field-

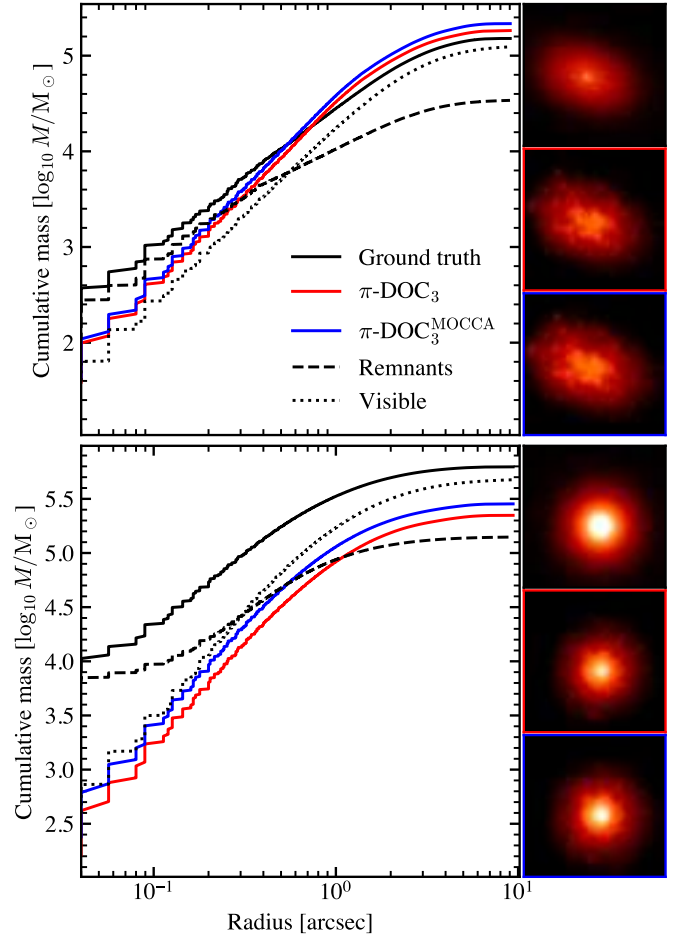


Fig. 14. Cumulative mass profiles. Top row: mock cluster (a) in Figure 4. Bottom row: mock cluster (d) in Figure 4. Solid black line shows the ground truth while the solid red and blue lines show the predicted mass maps from π -DOC₃ and π -DOC₃^{MOCCA}, respectively. Corresponding profiles for only remnants (dashed black line) and only visible mass (dotted black line) are also indicated, with 2D mass maps are shown on the right.

star decontamination, reconstructs the 2D mass distribution, and infers age, distance, ellipticity, and position angle, at ~ 10 inferences/s, processing the GCs within the combined PHAT and PHAST footprint in well under a minute — illustrating the scalability of this forward-modelling and deep learning approach for future surveys.

In synthetic tests representing a best-case scenario (see Section 4), π -DOC performs well. With the 2σ limit as a conservative confidence, we found that ages are recovered to within $\lesssim 3$ Gyr, with masses typically constrained to within a factor of 2-3. A systematic mass underestimation persists for the most massive clusters ($\gtrsim 10^6 M_{\odot}$), likely due to their limited training-set representation. Ellipticity and position angle are fairly well constrained for elliptical clusters, with a more significant scatter at low ellipticity ($\lesssim 0.1$); distance estimates remain comparatively uncertain ($\gtrsim 50$ kpc). Decontamination is most reliable at high S/N, but remains possible even at low S/N (suggesting π -DOC could be re-purposed for GC detection), though total magnitudes are systematically underestimated by up to ~ 1 mag, producing a non-negligible colour bias.

Applied to the PHAT and PHAST samples (see Section 6), π -DOC yields mass estimates broadly consistent with the literature, though systematically lower by a factor of 1.3–4 relative to Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024) — consistent with the high-mass underestimation seen in the synthetic tests. Interestingly, our results also suggests an average decrease in the GCs’ mass of around $1200 M_{\odot}$ per arcminute from the centre of M31 (see Section 6.2). This could be interpreted as a consequence of dynamical friction, but may also reflect the intrinsic biases and uncertainties of π -DOC.

The average distance estimate, 813 ± 31 kpc, is somewhat larger than typical M31 values (e.g. McConnachie et al. 2005), though consistent within uncertainties, while age estimates show the expected scatter from the age-metallicity degeneracy and the training set’s metal-poor composition, with better agreement at low metallicity. The ellipticity distribution agrees with Staneva et al. (1996) in aggregate, though not always cluster-by-cluster, likely reflecting known measurement biases between space- and ground-based imaging (Fréour et al. 2026). Together, these comparisons provide external validation of our approach.

We compared the latent-space representation of the observations (for mock testdata, see Section 5, for real data, see Section 6) to the training data via the Mahalanobis distance (D_M), finding that D_M typically correlates with agreement to independent studies and estimate reliability: the least similar clusters are systematically more massive, and especially more metal-rich, than the most similar ones — the latter plausibly linked to D_M ’s relationship with cluster colour. This reflects the limited training-set support for massive and metal-rich clusters, and identifies the model’s main current limitation, motivating more extensive simulation grids exploring this parameter space. It also suggests that colour and metallicity information already present in the network’s latent space could, with dedicated training, be exploited to jointly constrain age and metallicity in future extensions.

Preliminary tests also suggest π -DOC can be optimised for specific inference tasks (e.g. high-mass robustness, which plausibly could be extended to stellar remnant fraction detection, see Section 7), potentially reducing the need for very large training sets. A separate limitation is that π -DOC is currently tailored to HST semi-resolved GCs (F336W–F475W–F814W or F475W–F814W), restricting its applicability. We expect retraining to generalise well to other setups, though adapting across telescopes and host galaxies remains an open challenge; domain-adaptation techniques (Thuruthipilly et al. 2026) suggest very large survey-specific datasets may not be necessary. Future work should explore such techniques and a more flexible architecture (e.g. Parker et al. 2024, 2025), robust to passband combinations (e.g. via random band dropout) and image size, as required for more distant globular clusters.

Acknowledgements. This project was provided with computing resources by GENCI at IDRIS thanks to the following time allocations on the supercomputer Jean Zay (V100 partition): Grand Challenge-101470, A10-A0100412451, and A13-A0130412451. The authors would also like to acknowledge the High Performance Computing Center of the University of Strasbourg for supporting this work by providing access to computing resources. Part of the computing resources were funded by the Equipex Equip@Meso project (Programme Investissements d’Avenir) and the CPER Alsacalcul/Big Data. AM and PB acknowledge financial support by the IdEx framework of the University of Strasbourg. This work is partially supported by the grant GALBAR ANR-25-CE31-4684 of the French Agence Nationale de la Recherche.

References

Aarseth, S. J. 2003, *Gravitational N-Body Simulations*
 Abdi, H. & Williams, L. J. 2010, *WIREs Computational Statistics*, 2, 433

Adamo, A., Bradley, L. D., Vanzella, E., et al. 2024, *Nature*, 632, 513
 Alfaro-Cuello, M., Kacharov, N., Neumayer, N., et al. 2019, *ApJ*, 886, 57
 Arca sedda, M., Kamlah, A. W. H., Spurzem, R., et al. 2024, *MNRAS*, 528, 5140
 Askar, A., Askar, A., Pasquato, M., & Giersz, M. 2019, *MNRAS*, 485, 5345
 Askar, A., Vergara, M. C., & Ali, S. 2025, *arXiv e-prints*, arXiv:2510.03766
 Barbisan, E., Huang, J., Dage, K. C., et al. 2022, *MNRAS*, 514, 943
 Barmby, P. & Huchra, J. P. 2001, *The Astronomical Journal*, 122, 2458–2468
 Baumgardt, H. 2017, *MNRAS*, 464, 2174
 Baumgardt, H. & Makino, J. 2003, *MNRAS*, 340, 227
 Bellini, A., Bianchini, P., Varri, A. L., et al. 2017, *ApJ*, 844, 167
 Bialopetravičius, J. & Narbutis, D. 2020a, *A&A*, 633, A148
 Bialopetravičius, J. & Narbutis, D. 2020b, *AJ*, 160, 264
 Bialopetravičius, J., Narbutis, D., & Vansevičius, V. 2019, *A&A*, 621, A103
 Bianchini, P., Sills, A., & Miholics, M. 2017, *Monthly Notices of the Royal Astronomical Society*, 471, 1181
 Bianchini, P., Sills, A., van de Ven, G., & Sippel, A. C. 2017, *MNRAS*, 469, 4359
 Bianchini, P., van de Ven, G., Norris, M. A., Schinnerer, E., & Varri, A. L. 2016, *MNRAS*, 458, 3644
 Bianchini, P., Varri, A. L., Askar, A., Marklund, A., & Mastrobuono-Battisti, A. 2026, *ROLLIN’*: Rotating globular cluster simulations. I. The kinematic evolution of realistic direct N-body models
 Bissekennov, A., Pang, X., Kamlah, A., et al. 2025, *A&A*, 699, A196
 Boin, T., Casamiquela, L., Haywood, M., et al. 2026, *A&A*, 708, A215
 Bressan, A., Marigo, P., Girardi, L., et al. 2012, *MNRAS*, 427, 127
 Caldwell, N., Harding, P., Morrison, H., et al. 2009, *AJ*, 137, 94
 Chardin, J. & Bianchini, P. 2021, *MNRAS*, 504, 5656
 Chen, B., Liu, X., Xiang, M., et al. 2016, *AJ*, 152, 45
 Chen, Y. & Gnedin, O. Y. 2024, *The Open Journal of Astrophysics*, 7, 23
 Chen, Z., Williams, B., Lang, D., et al. 2025, *ApJ*, 979, 35
 Choi, J., Dotter, A., Conroy, C., et al. 2016, *The Astrophysical Journal*, 823, 102
 Conroy, C. & Gunn, J. E. 2010, *ApJ*, 712, 833
 Conroy, C., Gunn, J. E., & White, M. 2009, *ApJ*, 699, 486
 Dalcanton, J. J., Williams, B. F., Lang, D., et al. 2012, *ApJS*, 200, 18
 Dold, D. & Fahrion, K. 2022, *Astronomy & Astrophysics*, 663, A81
 Dotter, A., Sarajedini, A., Anderson, J., et al. 2010, *ApJ*, 708, 698
 Euclid Collaboration, Scaramella, R., Amiaux, J., et al. 2022, *A&A*, 662, A112
 Euclid Collaboration, Voggel, K., Lancon, A., et al. 2025, *A&A*, 693, A251
 Fan, Z. & de Grijs, R. 2012, *MNRAS*, 424, 2009
 Fan, Z., de Grijs, R., & Zhou, X. 2010, *ApJ*, 725, 200
 Forbes, D. A. 2020, *MNRAS*, 493, 847
 Forbes, D. A., Bastian, N., Gieles, M., et al. 2018, *Proceedings of the Royal Society of London Series A*, 474, 20170616
 Forbes, D. A. & Bridges, T. 2010, *MNRAS*, 404, 1203
 Fréour, L., Leitinger, E., Pancino, E., Zocchi, A., & van de Ven, G. 2026, *arXiv e-prints*, arXiv:2603.08432
 Gal, Y. & Ghahramani, Z. 2015a, *arXiv e-prints*, arXiv:1506.02157
 Gal, Y. & Ghahramani, Z. 2015b, *arXiv e-prints*, arXiv:1506.02142
 Gal, Y., Islam, R., & Ghahramani, Z. 2017, *arXiv e-prints*, arXiv:1703.02910
 Gardner, J. P., Mather, J. C., Abbott, R., et al. 2023, *PASP*, 135, 068001
 Gieles, M., Heggie, D. C., & Zhao, H. 2011, *MNRAS*, 413, 2509
 Giersz, M., Heggie, D. C., Hurley, J. R., & Hypki, A. 2013, *Monthly Notices of the Royal Astronomical Society*, 431, 2184
 Gonzalez, C., Gotkowsky, K., Fuchs, M., et al. 2022, *Distance-based detection of out-of-distribution silent failures for Covid-19 lung lesion segmentation*
 Green, G. 2018, *The Journal of Open Source Software*, 3, 695
 Guiglion, G., Nepal, S., Chiappini, C., et al. 2024, *A&A*, 682, A9
 Harris, W. E. 2010, *arXiv e-prints*, arXiv:1012.3224
 Hiegel, J., Thélie, É., Aubert, D., et al. 2023, *A&A*, 679, A125
 Holland, S. 1998, *AJ*, 115, 1916
 Hurley, J. R., Pols, O. R., & Tout, C. A. 2000, *MNRAS*, 315, 543
 Hurley, J. R., Tout, C. A., & Pols, O. R. 2002, *MNRAS*, 329, 897
 Huxor, A. P., Mackey, A. D., Ferguson, A. M. N., et al. 2014, *MNRAS*, 442, 2165
 Hypki, A. & Giersz, M. 2013, *MNRAS*, 429, 1221
 Ibata, R. A., Gilmore, G., & Irwin, M. J. 1995, *MNRAS*, 277, 781
 Ivezić, Ž., Kahn, S. M., Tyson, J. A., et al. 2019, *ApJ*, 873, 111
 Jindal, A., Webb, J. J., & Bovy, J. 2019, *MNRAS*, 487, 3693
 Kamlah, A. W. H., Leveque, A., Spurzem, R., et al. 2022, *MNRAS*, 511, 4060
 Kang, K., Setlur, A., Tomlin, C., & Levine, S. 2024, *Deep Neural Networks Tend To Extrapolate Predictably*
 Kingma, D. P. & Ba, J. 2014, *arXiv e-prints*, arXiv:1412.6980
 Kruijssen, J. M. D. 2015, *Monthly Notices of the Royal Astronomical Society*, 454, 1658
 Kuzma, P. B., Da Costa, G. S., Mackey, A. D., & Roderick, T. A. 2016, *MNRAS*, 461, 3639
 Lahén, N., Naab, T., Johansson, P. H., et al. 2020, *ApJ*, 891, 2
 Lahén, N., Rantala, A., Naab, T., et al. 2025, *MNRAS*, 538, 2129
 Leaman, R., VandenBerg, D. A., & Mendel, J. T. 2013, *Monthly Notices of the Royal Astronomical Society*, 436, 122

- Leaman, R., VandenBerg, D. A., & Mendel, J. T. 2013, *MNRAS*, 436, 122
- Lee, K., Lee, K., Lee, H., & Shin, J. 2018, A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks
- Lejeune, T., Cuisinier, F., & Buser, R. 1997, *A&AS*, 125, 229
- Lejeune, T., Cuisinier, F., & Buser, R. 1998, *A&AS*, 130, 65
- Liu, X.-W., Zhao, G., & Hou, J.-L. 2015, *Research in Astronomy and Astrophysics*, 15, 1089
- Malhan, K., Ibata, R. A., & Martin, N. F. 2018, *MNRAS*, 481, 3442
- Maraston, C. 2005, *MNRAS*, 362, 799
- Marklund, A., Bianchini, P., Varri, A. L., Kraljic, K., & Pagnini, G. 2026, arXiv e-prints, arXiv:2607.29499
- Massari, D., Koppelman, H. H., & Helmi, A. 2019, *A&A*, 630, L4
- McConnachie, A. W., Irwin, M. J., Ferguson, A. M. N., et al. 2005, *MNRAS*, 356, 979
- Miholics, M., Webb, J. J., & Sills, A. 2016, *MNRAS*, 456, 240
- Mosby, G., Rauscher, B. J., Bennett, C., et al. 2020, *Journal of Astronomical Telescopes, Instruments, and Systems*, 6, 046001
- Mowla, L., Iyer, K., Asada, Y., et al. 2024, *Nature*, 636, 332
- Myeong, G. C., Vasiliev, E., Iorio, G., Evans, N. W., & Belokurov, V. 2019, *MNRAS*, 488, 1235
- Pagnini, G., Di Matteo, P., Haywood, M., et al. 2026, *A&A*, 708, A161
- Pagnini, G., Di Matteo, P., Haywood, M., et al. 2025, *A&A*, 693, A155
- Parker, L., Lanusse, F., Golkar, S., et al. 2024, *Monthly Notices of the Royal Astronomical Society*, 531, 4990–5011
- Parker, L., Lanusse, F., Shen, J., et al. 2025, AION-1: Omnimodal Foundation Model for Astronomical Sciences
- Pasquato, M., & Chung, C. 2016, *A&A*, 589, A95
- Pasquato, M., Trevisan, P., Askar, A., et al. 2024, *ApJ*, 965, 89
- Peacock, M. B., Maccarone, T. J., Knigge, C., et al. 2010a, *MNRAS*, 402, 803
- Peacock, M. B., Maccarone, T. J., Knigge, C., et al. 2010b, *VizieR Online Data Catalog: M31 globular cluster system (Peacock+, 2010)*, *VizieR Online Data Catalog: J/MNRAS/402/803*. Originally published in: 2010MNRAS.402..803P
- Pechetti, R., Seth, A., Kamann, S., et al. 2022, *ApJ*, 924, 48
- Pfeffer, J., Lardo, C., Bastian, N., Saracino, S., & Kamann, S. 2021, *MNRAS*, 500, 2514
- Renaud, F., Agertz, O., & Gieles, M. 2017, *MNRAS*, 465, 3622
- Renaud, F., Gieles, M., & Boily, C. M. 2011, *MNRAS*, 418, 759
- Rodriguez, C. L., Morscher, M., Wang, L., et al. 2016, *MNRAS*, 463, 2109
- Ronneberger, O., Fischer, P., & Brox, T. 2015, arXiv e-prints, arXiv:1505.04597
- Rosenberg, A., Saviane, I., Piotto, G., & Aparicio, A. 1999, *AJ*, 118, 2306
- Sakari, C. M., Shetrone, M. D., McWilliam, A., & Wallerstein, G. 2021, *MNRAS*, 502, 5745
- Sakari, C. M. & Wallerstein, G. 2016, *MNRAS*, 456, 831
- Sakari, C. M. & Wallerstein, G. 2022, *MNRAS*, 512, 4819
- Schlegel, D. J., Finkbeiner, D. P., & Davis, M. 1998, *ApJ*, 500, 525
- Singlow, T., Techa-Angkoon, P., Bootkrajang, J., Suwannajak, C., & Tanakul, N. 2022, in 2022 19th International Conference on Electrical Engineering/Electronics, 138
- Sollima, A., Dalessandro, E., Beccari, G., & Pallanca, C. 2017, *Monthly Notices of the Royal Astronomical Society*, 464, 3871
- Spitzer, L. 1987, *Dynamical evolution of globular clusters*
- Staneva, A., Spassova, N., & Golev, V. 1996, *A&AS*, 116, 447
- Taylor, E. D., Read, J. I., Orkney, M. D. A., et al. 2025, *Nature*, 645, 327
- Thuruthipilly, H., Lisiecki, K., Junaes, et al. 2026, *A&A*, 711, A187
- Trenti, M., & van der Marel, R. 2013, *MNRAS*, 435, 3272
- Usher, C., Brodie, J. P., Forbes, D. A., et al. 2019, *MNRAS*, 490, 491
- Usher, C., Caldwell, N., & Cabrera-Ziri, I. 2024, *MNRAS*, 528, 6010
- Valcin, D., Jimenez, R., Lardo, C., Seljak, U., & Verde, L. 2026, arXiv e-prints, arXiv:2603.04872
- Valcin, D., Jimenez, R., Seljak, U., & Verde, L. 2025, *J. Cosmology Astropart. Phys.*, 2025, 030
- Vanzella, E., Claeysens, A., Welch, B., et al. 2023, *ApJ*, 945, 53
- Vesperini, E., & Heggie, D. C. 1997, *MNRAS*, 289, 898
- Viaña, J., Lee, J. C., Vanderburg, A., et al. 2026, *ApJ*, 1004, 8
- Wang, L., Spurzem, R., Aarseth, S., et al. 2016, *Monthly Notices of the Royal Astronomical Society*, 458, 1450–1465
- Wang, L., Spurzem, R., Aarseth, S., et al. 2015, *MNRAS*, 450, 4070
- Watkins, L. L., van der Marel, R. P., Bellini, A., & Anderson, J. 2015, *ApJ*, 803, 29
- Webb, J. J., Reina-Campos, M., & Kruijssen, J. M. D. 2024, *ApJ*, 975, 242
- Webb, J. J. & Vesperini, E. 2016, *MNRAS*, 463, 2383
- Westera, P., Lejeune, T., Buser, R., Cuisinier, F., & Bruzual, G. 2002, *A&A*, 381, 524
- Williams, B. F., Durbin, M., Lang, D., et al. 2023, *ApJS*, 268, 48
- Williams, B. F., Lang, D., Dalcanton, J. J., et al. 2014, *ApJS*, 215, 9
- Worthey, G. 1994, *ApJS*, 95, 107
- Worthey, G. 1999, in *Astronomical Society of the Pacific Conference Series*, Vol. 192, *Spectrophotometric Dating of Stars and Galaxies*, ed. I. Hubeny, S. Heap, & R. Cornett, 283
- Wu, K., Tanikawa, A., Flammini Dotti, F., et al. 2026, *A&A*, 709, A142
- York, D. G., Adelman, J., Anderson, Jr., J. E., et al. 2000, *AJ*, 120, 1579
- Yuan, H.-B., Liu, X.-W., Huo, Z.-Y., et al. 2015, *Monthly Notices of the Royal Astronomical Society*, 448, 855
- Zhang, B., Chen, B., Yuan, H., et al. 2025, *ApJS*, 278, 16
- Zhao, Y., Askar, A., Lu, Y., et al. 2026, *ApJ*, 998, 328
- Zoccali, M., Renzini, A., Ortolani, S., et al. 2003, *A&A*, 399, 931
- Zocchi, A., Gieles, M., & Hénault-Brunet, V. 2017, *MNRAS*, 468, 4429

Appendix A: Realism of mock images

Fig. A.1 shows a qualitative comparison between five mock images (top row) and five GCs from the PHAT survey (middle row). The bottom row shows the corresponding pixel-CMDs.

Appendix B: π -DOC in two versus three passbands

We use two versions of π -DOC for the survey analyses: π -DOC₃ for PHAT and π -DOC₂ for PHAST, the latter being limited to two passbands. To assess the effect of the missing F336W passband, we directly compare the two networks and their predictions for the mock images in our test set. Figure B.1 shows that the networks generally agree within the typical confidence limits (Section 4.2).

A similar comparison for the PHAT GCs reveals the same overall trends. In general, the two networks agree reasonably well for each quantity, although with a non-negligible scatter. The mean absolute discrepancies (differences computed as π -DOC₂ minus π -DOC₃) are approximately 0.28 dex for the luminosity in both passbands (F475W and F814W), 0.24 dex for mass, 1.3 Gyr for age, 23.5 kpc for distance, 0.02 for ellipticity, and 66° for position angle. Agreement is particularly good for age across the full range of values. For luminosity, however, π -DOC₃ tends to predict brighter values for the faintest objects, suggesting that the third passband is especially informative for low-luminosity GCs; note that this regime is close to the edge of the coverage by the mock test data. At the bright and massive ends, the opposite trend is seen, with π -DOC₂ typically predicting luminosities and masses a factor ~ 1.5 higher than π -DOC₃. Distances agree well overall, although π -DOC₃ more often predicts shorter distances, possibly because the additional pixel CMDs improve the distance inference. Ellipticity is concentrated near $e \sim 0.1$, indicating that it is only weakly constrained; this is consistent with the synthetic tests, where nearly spherical clusters tend to be overestimated and objects with $e < 0.1$ show substantial scatter. Position angle shows the largest discrepancies, likely because most clusters in the sample are only mildly elliptical. By contrast, the synthetic tests in Section 4 showed that more elliptical clusters ($e \gtrsim 0.1$) are much better constrained, implying that such objects would likely have produced a clearer signal in the real-sample predictions.

Appendix C: Robustness to noise

Most of our clusters are faint, whereas most of the backgrounds are, on average, a few magnitudes brighter. The accuracy of the predictions should therefore depend on the relative brightness of the GC and the background. In Fig. C.1, we show the two-dimensional distributions of the errors for each quantity as a function of the magnitude difference between the GC and the corresponding background in the F814W passband²². Negative values indicate that the GC is brighter, while positive values indicate that the background is brighter. Qualitatively, we find that the errors depend on the relative GC-to-background brightness in much the same way as they do on GC brightness alone (see Fig. 6): fainter clusters generally correspond to lower S/N, and the spread in the predictions increases substantially once the background dominates.

The top row shows the magnitudes and total mass, both derived from their respective maps. At low S/N, there is a significant bias in each quantity toward underestimating brightness and

total mass. Even at high S/N, the GC brightness remains systematically underestimated in F336W and F475W, with the smallest bias occurring near the most common GC-to-background brightness difference in the training data; the bias increases again toward both the lowest and highest S/N. For F814W the bias is instead smallest at high S/N and largest at low S/N. The origin of this passband-dependent asymmetry is not clear from our analysis. The pattern is nevertheless suggestive of a colour-dependent bias in the decontamination, possibly linked to how the network blends GC and background differently across passbands at high relative brightness. Regardless, decontaminated clusters and their integrated magnitudes should be regarded as upper limits, since the true clusters are in most cases brighter than predicted by π -DOC.

The total mass shows no such systematic bias at high S/N, but is instead associated with a larger absolute spread, of up to a factor of $\sim 3 - 4$. The regression-head outputs (bottom row) behave oppositely: they remain broadly symmetric around zero and are therefore unbiased. Only the ellipticity estimates show a bias to be underestimated in background-dominated images, where the stellar distribution becomes more diffuse and the cluster consequently appears more circular (the central regions of GCs are typically more circular; see Marklund et al. 2026). This effect is at most 0.03, small compared with the overall spread. Moreover, for GCs that are brighter than the background, the estimates are much better constrained, which is especially clear for age and ellipticity.

The low-S/N mock images in Fig. 4 already showed that the network is reasonably robust to noise (although π -DOC's estimates were less accurate than the corresponding high-S/N examples). To explore this further, we investigate how the predictions change when the same mock image of a relatively faint cluster (F814W ~ 17.2) is combined with backgrounds of different magnitudes in Fig. C.2.

Most predictions remain relatively accurate and scatter around the ground truth, the magnitudes are in all cases except one overestimated, as expected from the discussion above (see Fig. C.1). Overall, the predictions remain fairly robust to the varying noise levels. The total mass varies within a factor of ~ 1.8 , the position angle within $\sim 10^\circ$, the ellipticity within 0.09 (lower predicted values for more diffuse distributions where the GC inevitably appears more circular), the distance within ~ 50 kpc, the age within 6 Gyr (with the background-free and the faintest background lying within 0.8 Gyr to the ground truth), and the integrated magnitude within 0.6 mag in F814W. Except for the age, these variations lie within the confidence range previously defined for each quantity in Section 4.2, and also show that more reliable estimates can be expected for GCs in less crowded and less bright regions.

Appendix D: Epistemic Uncertainty

For each input image, we perform 20 stochastic forward passes (using an MC dropout rate of 0.1) and compute the mean and standard deviation of the resulting predictions. We adopt the mean prediction as the final model output, while the standard deviation is used to quantify the model's confidence. This procedure naturally yields pixel-wise uncertainty maps for spatially resolved outputs (e.g. mass and magnitude maps), as well as uncertainties for our scalar regression targets: age, distance, ellipticity, and position angle. Because we predict the $\cos(2P.A.)$ and $\sin(2P.A.)$ components rather than the position angle directly, and to account for the circular nature of angular quantities, the epistemic uncertainty is estimated using the circular

²² We note that there are no substantial differences for the F336W and F475W passbands.

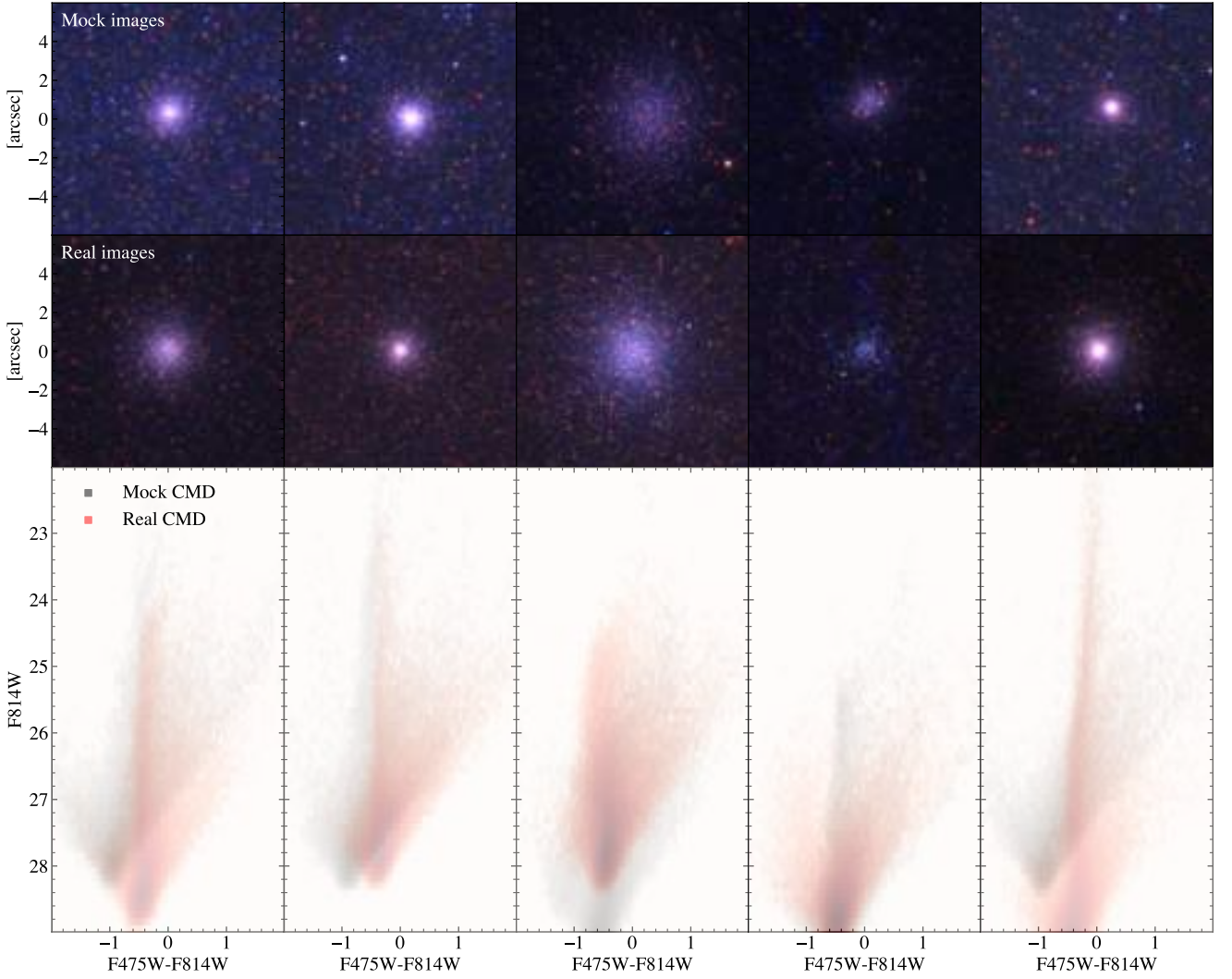


Fig. A.1. Visual comparison of mock images (top row), real images (middle row), and the corresponding pixel-CMDs (bottom row) for M31 GCs. The clusters resemble each other in appearance: similar sizes, shapes, and brightness. Mock images were chosen based on their visual resemblance, and may not correspond to physical similarity. There are notable differences between the images, especially noticeable in the pixel-CMDs, for example with respect to the base level and colour. Each RGB image is scaled identically to the inputs in Figure 4.

standard deviation. Specifically, we convert each sampled angle to its unit-vector representation ($\cos \text{P.A.}$, $\sin \text{P.A.}$), compute the mean resultant vector over stochastic forward passes, and derive the circular standard deviation as $\sigma_{\text{circ}} = \sqrt{-2 \ln R}$, where R is the magnitude of the mean resultant vector. The mean position angle is obtained from the averaged P.A. = $\frac{1}{2} \arctan 2(\sin(2\text{P.A.}), \cos(2\text{P.A.}))$.

The resulting epistemic uncertainties are shown in Fig. D.1 against the absolute error of the predictions for the test set. We see that the epistemic uncertainties, for all predicted quantities, significantly underestimates the true error. As such, it does not tell us much about the physical uncertainty in the prediction of a cluster, but rather indicates the confidence level of π -DOC's prediction.

Appendix E: Catalogue of properties for GCs in M31

π -DOC's estimates for the various quantities discussed throughout this paper are indicated in Table E.1 for a subset of the GCs in the PHAT and PHAST footprints.

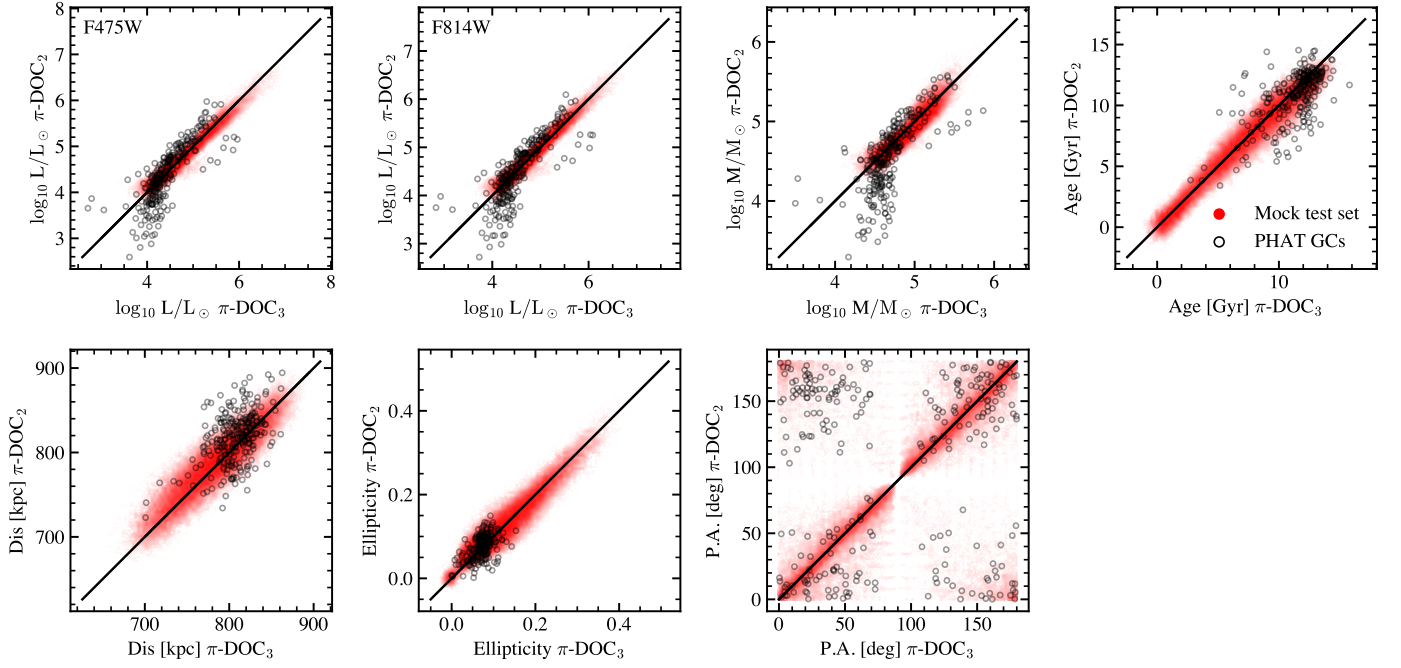


Fig. B.1. Comparisons of estimates between π -DOC₃ (x-axis) and π -DOC₂ (y-axis) for the total luminosities (within FoV) in F475W and F814W passbands, total mass (within FoV), age, distance, ellipticity, and position angle. Overall, the networks agree (black line indicates a 1:1 correspondence), however, they show some intrinsic scatter with respect to each other for the mock test set (red points). This scatter is also seen for the estimates of M31's GC properties which otherwise follow the distributions of the test set.

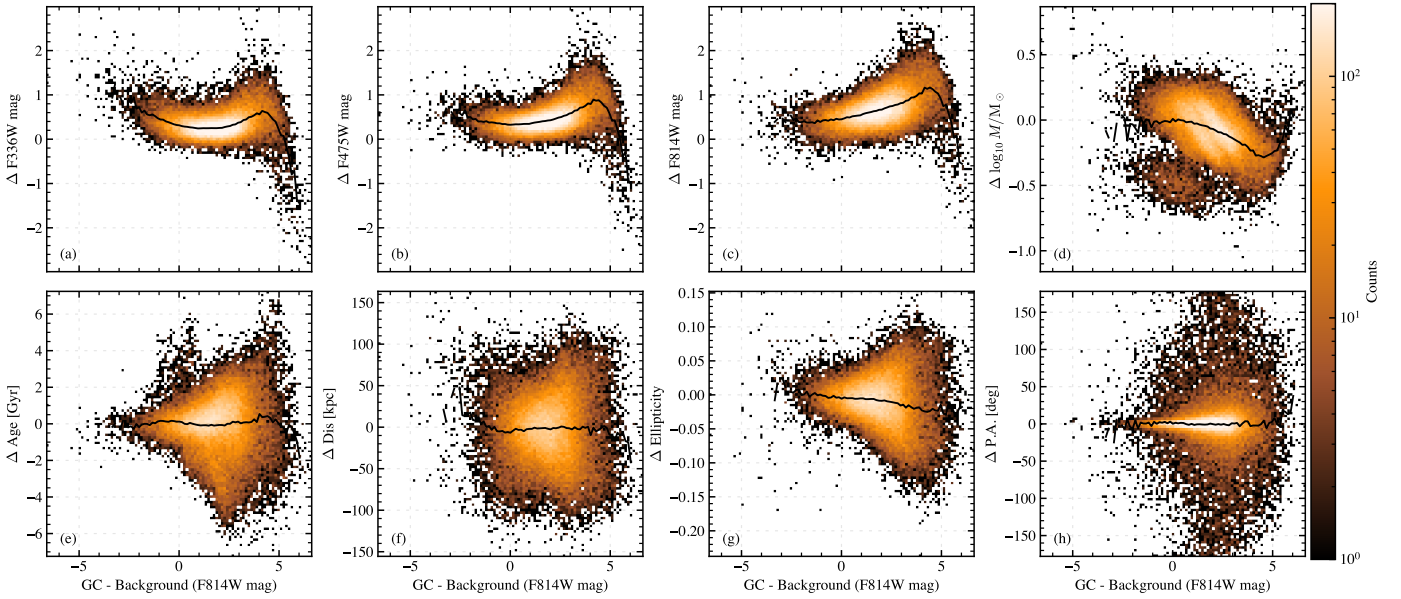


Fig. C.1. Difference between predicted value and ground truth for every quantity and how they depend on the relative brightness between a cluster and its associated background. π -DOC is less accurate for background dominated images, indicating that low S/N, in particular, is a limitation. The solid black line indicates a moving mean.

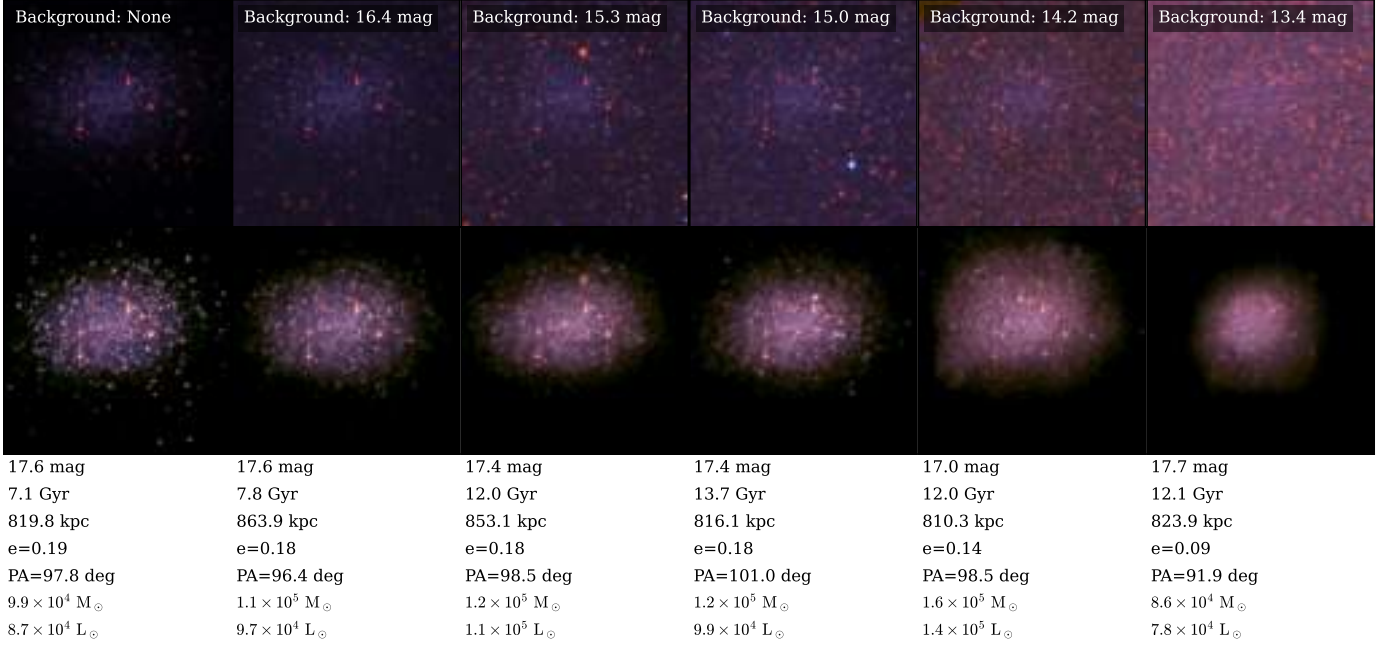


Fig. C.2. One mock GC combined with different backgrounds (top row) scaled identically as the inputs in Figure 4, and π -DOC’s predictions (bottom row), scaled identically as the decontaminated images in Figure 4. Each mock image’s associated predictions are indicated beneath the decontaminated image. The ground truth, in the same order as the predictions from top to bottom, are: $1.4 \times 10^5 M_{\odot}$, $1.3 \times 10^5 L_{\odot}$, $\phi = 97.9^{\circ}$, $e = 0.20$, 842.8 kpc, 8.0 Gyr, and the total integrated magnitude in the F814W passband is 17.2 mag. The predictions remain within the corresponding confidence limits, and the decontamination is fairly robust to decreasing S/N.

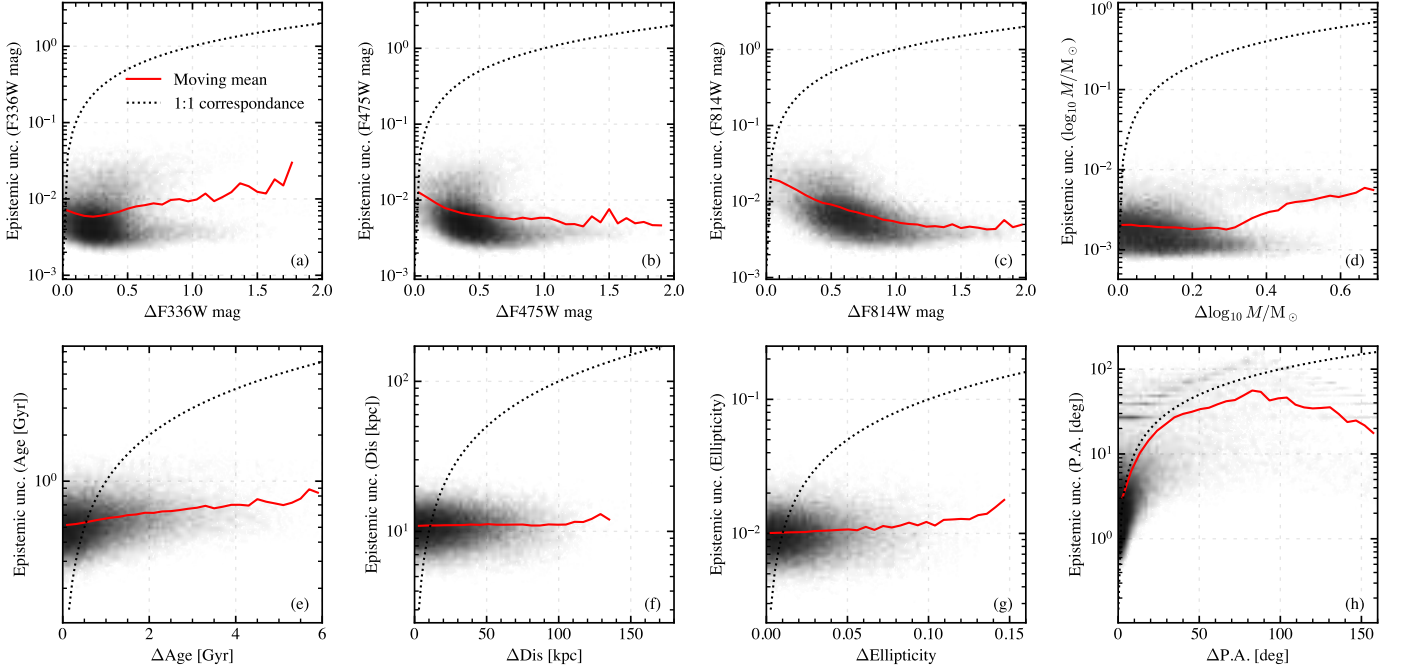


Fig. D.1. Epistemic uncertainty compared to actual errors for each predicted quantity. In all cases, the epistemic uncertainty significantly underestimates the error.

Table E.1. Globular cluster catalogue.

Name	Ra deg	Dec deg	F336W mag	F475W mag	F814W mag	log $M/M_{\odot}$ dex	Age Gyr	Distance kpc	Ellipticity	P.A. deg	D_M
SK122C	10.82	41.51	20.09 ± 0.00	19.07 ± 0.00	19.27 ± 0.01	4.54 ± 0.00	13.08 ± 0.55	815.84 ± 11.36	0.07 ± 0.01	131.47 ± 38.04	16.30
ACH 12	10.76	41.21	19.75 ± 0.02	18.68 ± 0.02	18.92 ± 0.01	4.61 ± 0.00	12.84 ± 0.55	802.16 ± 7.13	0.08 ± 0.01	41.98 ± 14.50	20.66
NB35-AU4	10.64	41.31	19.61 ± 0.01	18.61 ± 0.01	18.91 ± 0.01	4.60 ± 0.00	12.12 ± 0.38	799.15 ± 17.70	0.08 ± 0.01	47.21 ± 51.46	24.02
NB21-AU5	10.66	41.27	16.56 ± 0.00	15.49 ± 0.00	15.73 ± 0.00	5.46 ± 0.00	11.97 ± 0.65	808.55 ± 7.18	0.06 ± 0.01	66.99 ± 10.04	19.29
V129-BA4	11.44	41.87	18.41 ± 0.01	17.40 ± 0.01	17.75 ± 0.01	4.86 ± 0.00	9.81 ± 0.66	778.87 ± 10.40	0.04 ± 0.01	59.02 ± 5.85	16.71
B103-G165	10.62	41.30	17.18 ± 0.02	16.05 ± 0.02	16.16 ± 0.02	5.20 ± 0.00	7.86 ± 1.00	779.56 ± 9.35	0.05 ± 0.01	9.81 ± 6.93	22.72
B104-NB5	10.62	41.29	18.43 ± 0.02	17.38 ± 0.02	17.65 ± 0.03	4.83 ± 0.01	12.08 ± 0.99	808.71 ± 13.82	0.04 ± 0.02	15.64 ± 77.88	23.97
B107-G169	10.63	41.33	17.24 ± 0.01	16.13 ± 0.01	16.35 ± 0.01	5.10 ± 0.00	7.72 ± 0.82	801.63 ± 14.32	0.04 ± 0.01	19.32 ± 59.76	19.23
B112-G174	10.64	41.30	18.76 ± 0.03	17.71 ± 0.03	17.92 ± 0.03	4.78 ± 0.01	9.19 ± 0.89	790.40 ± 15.85	0.07 ± 0.02	53.23 ± 8.36	24.06
B115-G177	10.64	41.23	18.34 ± 0.02	17.18 ± 0.02	17.35 ± 0.02	4.91 ± 0.00	8.41 ± 0.99	821.37 ± 14.77	0.11 ± 0.01	178.59 ± 10.88	23.06
B119-NB14	10.65	41.29	18.71 ± 0.02	17.65 ± 0.02	17.90 ± 0.03	4.79 ± 0.00	12.38 ± 1.05	815.45 ± 15.68	0.06 ± 0.03	135.05 ± 109.79	23.88
B124-NB10	10.67	41.26	15.73 ± 0.01	14.61 ± 0.01	14.81 ± 0.01	5.68 ± 0.00	7.29 ± 0.61	813.92 ± 12.60	0.09 ± 0.01	64.34 ± 7.37	22.98
B126-G184	10.68	41.21	18.46 ± 0.01	17.45 ± 0.01	17.77 ± 0.01	4.79 ± 0.00	13.30 ± 1.63	832.34 ± 12.32	0.03 ± 0.02	164.14 ± 7.77	23.92
B129	10.70	41.42	19.77 ± 0.02	18.48 ± 0.02	18.44 ± 0.02	4.69 ± 0.00	10.45 ± 0.82	814.72 ± 16.68	0.09 ± 0.02	42.53 ± 55.13	16.73
B131-G189	10.71	41.29	16.11 ± 0.01	15.07 ± 0.01	15.37 ± 0.02	5.45 ± 0.00	8.83 ± 1.01	821.85 ± 15.59	0.03 ± 0.01	37.90 ± 6.34	19.90
B132	10.71	41.26	17.36 ± 0.02	16.32 ± 0.02	16.57 ± 0.02	5.18 ± 0.00	12.19 ± 1.32	822.85 ± 8.89	0.08 ± 0.02	166.57 ± 57.94	19.38
B133	10.71	41.39	19.56 ± 0.00	18.78 ± 0.00	19.27 ± 0.00	4.45 ± 0.00	9.01 ± 0.67	815.29 ± 11.43	0.07 ± 0.01	0.75 ± 89.15	17.97
B134-G190	10.72	41.23	17.90 ± 0.02	16.86 ± 0.02	17.12 ± 0.01	4.90 ± 0.00	11.07 ± 0.90	828.48 ± 10.07	0.03 ± 0.02	160.92 ± 10.22	25.17
B136-G194	10.72	41.33	18.29 ± 0.02	17.27 ± 0.02	17.54 ± 0.02	4.94 ± 0.00	11.77 ± 0.72	817.77 ± 11.63	0.02 ± 0.01	10.17 ± 56.64	24.36
B138	10.73	41.31	18.75 ± 0.02	17.70 ± 0.02	17.97 ± 0.02	4.80 ± 0.00	11.54 ± 1.29	813.50 ± 13.64	0.07 ± 0.01	167.58 ± 63.68	23.20
...	...	...	...	...	...	...	...	...	...	...	...
SK086B	10.43	40.89	N/A	19.90 ± 0.04	20.68 ± 0.04	3.77 ± 0.01	13.94 ± 0.94	778.57 ± 11.64	0.06 ± 0.01	146.58 ± 6.12	14.63
SK107B	10.64	41.18	N/A	19.53 ± 0.01	19.79 ± 0.01	4.39 ± 0.00	11.44 ± 0.93	844.42 ± 12.87	0.08 ± 0.02	163.26 ± 2.71	21.59
V014	10.31	40.57	N/A	17.95 ± 0.02	18.49 ± 0.02	4.61 ± 0.00	7.07 ± 0.85	823.28 ± 13.84	0.09 ± 0.02	174.57 ± 14.02	9.41
V202	10.20	40.93	N/A	19.27 ± 0.02	19.68 ± 0.02	4.27 ± 0.00	12.55 ± 0.63	820.29 ± 11.02	0.08 ± 0.02	7.49 ± 51.13	15.66
WH2	10.00	40.56	N/A	21.42 ± 0.01	21.77 ± 0.01	3.63 ± 0.01	12.51 ± 0.76	804.16 ± 14.18	0.09 ± 0.01	173.43 ± 6.24	10.04

Notes. Names, coordinates, and π -DOC's estimates for various quantities: total integrated magnitudes in the F336W, F475W, and F814W, passbands that are derived from the decontaminated images, total mass derived from mass map, age, distance, ellipticity, position angle, and the Mahalanobis distance to the training manifold. Where applicable, columns also indicate π -DOC's epistemic uncertainties (for physical uncertainties we refer to our 2σ confidence limits). The epistemic uncertainties are provided as separate columns in the online version of the table. GCs in the PHAST survey have no F336W magnitude counterpart and their estimates are inferred from π -DOC₂.